

Market Order Mechanisms*

Benjamin Brooks Songzi Du Linchen Zhang

May 27, 2026

Abstract

A homogeneous good is for sale to a large number of buyers with multi-unit demand. There are both common and idiosyncratic components to preferences. Trade occurs via a *market order mechanism*: Each buyer places an order for a number of units; the orders are filled in a random sequence; and buyers pay a price per unit that is a function of the aggregate order. We study the limit of such economies as the numbers of buyers and units grows large. If the pricing rules have vanishing *price impact* and *span*, then regardless of the sequence of information structures and equilibria, the limiting market outcome is guaranteed to be *competitive*: The goods are sold to the buyers with the highest values, at an expected price that is equal to the expected marginal value for the good. It is impossible to achieve similar guarantees for information aggregation.

KEYWORDS: Information aggregation, market order, limit order, large market, competition, revenue guarantee.

JEL CLASSIFICATION: C72, D44, D82, D83.

*Brooks: Department of Economics, University of Chicago, babrooks@uchicago.edu; Du: Department of Economics, University of California San Diego, sodu@ucsd.edu; Zhang: Department of Economics, University of California San Diego, liz032@ucsd.edu. We are very grateful for comments from Piotr Dworczak, Ilan Kremer, Doron Ravid, Phil Reny, and numerous seminar and conference audiences. Previous versions of this paper were circulated as “An Informationally-Robust Market Model of Perfect Competition” and “Participation Adaptive Pricing.” This research has been supported by NSF #2215259 and #2215475.

1 Introduction

1.1 Strategic Trade in Large Markets

A large and influential literature, going back to Wilson (1977), has studied models of strategic trade in large markets with private information. While a handful of papers have analyzed two-sided markets, most of the literature concerns one-sided markets in which only the buyers of the good have private information. The literature has primarily focused on what we will refer to as *limit order mechanisms*:¹ Each prospective buyer's action represents a limit on the prices at which they are willing to trade a given quantity. Prominent examples are discriminatory and uniform price auctions. For such trading protocols, the literature has described conditions on the buyers' information such that when the population grows without bound, there are sequences of equilibria that are *competitive in the large*: in the limit, the goods sell to the buyers with the highest values, and the marginal buyer earns no rents. Moreover, in these equilibria, *information is aggregated* through prices: the prices paid for the good converge to the marginal buyer's willingness to pay. The signals that buyers may observe are usually assumed to be ordered and the equilibrium strategies are monotone, with further assumptions on information and preferences being used to support the monotone strategies as an equilibrium. At the same time, the literature has given a number of examples of economies that violate these hypotheses, and in which the equilibrium outcome is not close to competitive, no matter how large the market is.² Needless to say, for these examples, information is also not aggregated.

There is an inherent tension in trying to achieve a competitive outcome through information aggregation. In order for prices to be close to values, buyers must have a relatively rich set of actions they can take, so that through their behavior they can effectively communicate their perceived values to the market. Such is the case with limit order mechanisms, wherein buyers express in fine detail how much they are willing to pay. But a rich set of actions inevitably also creates rich opportunities to manipulate the market, which can sometimes lead to prices being systematically below values, and therefore not competitive. It is natural to ask whether there are alternative market structures, in which buyers can only communicate coarser information to the market, which while precluding information aggregation might end up facilitating greater competition.

¹See, e.g., Milgrom (1979, 1981), Pesendorfer and Swinkels (1997, 2000), Kremer (2002), Bali and Jackson (2002), Lauermaun and Wolinsky (2017), and Lauermaun and Speit (2023).

²See, e.g., Milgrom (1979), Engelbrecht-Wiggans et al. (1983), Bali and Jackson (2002), Bergemann, Brooks, and Morris (2017), and Barelli, Govindan, and Wilson (2024).

In this paper, we study such a class of trading protocols that we call *market order mechanisms*: Each buyer places an order for a number of units; the orders are filled in a random sequence; and buyers pay a price per unit that depends on the aggregate order. This pricing rule parametrizes the market order mechanism. We consider sequences of economies where the number of buyers and the number of units for sale both grow in proportion, and the pricing rule also varies with the size of the economy. Each buyer may demand multiple units, with downward sloping demand. The marginal willingness to pay for each unit depends on a component which is common to all buyers as well as an idiosyncratic component. Marginal values are assumed to be increasing in both the common and idiosyncratic components. Each buyer is assumed to know their idiosyncratic components of their values for each marginal unit. Like the preceding literature, we assume that there is very little uncertainty about the empirical distribution of the idiosyncratic components when the market is large, as in Pesendorfer and Swinkels (2000), for example. No further assumptions are made on the distribution of buyers' preferences, on what buyers know about the common component, others' idiosyncratic components, and others' information. In particular, the signals observed by the buyers are not assumed to have any order structure, and no assumptions are made about the buyers' strategies, except that they comprise a Bayes Nash equilibrium.

1.2 Overview of Results

Our main result consists of two conditions on the sequence of pricing rules such that the equilibrium outcome is guaranteed to be competitive in the large. This conclusion holds, no matter what is the particular sequence of information structures and Bayes Nash equilibria. Thus, in contrast to the case with limit order mechanisms—where competitive outcomes only obtain for certain sequences of distributions, information structures, and equilibria—market order mechanisms with such pricing rules are guaranteed to produce competitive outcomes without any further assumptions about the information and equilibrium.

We now describe the critical conditions on the sequence of pricing rules. One way in which the market might fail to be close to competitive is that the market price is significantly lower than the value of the marginal non-buyer, but if that buyer were to purchase, the price would jump up above their value. Our first condition rules this out. In particular, the *price impact* is the maximum amount by which the price changes in response to one additional order. For the outcome of the market to be competitive in the large, the price impact must vanish as the economy grows large.

Another way in which the market might fail to be competitive in the large is if the information of non-buyers about aggregate demand is systematically different from that

of buyers. In that case, it might be that the interim expected market price is somewhere in between the respective values of the marginal buyer and the marginal non-buyer. The second condition is a property of the pricing rule that rules out certain kinds of systematic differences in the information of buyers and non-buyers. In particular, we define the *span* to be the range of aggregate orders over which the pricing rule takes on intermediate values. For purely expositional convenience, we define the span in normalized units of the aggregate order, divided by the maximum total order in the economy. For aggregate orders below the span, the price is low enough that the good would certainly sell out, and for aggregate orders above the span, the price is sufficiently high that no one would want to buy. Our proof shows that equilibrium revenue is minimized when, with very high probability, the aggregate order is within the span. When the span is negligible in size, there must be almost no uncertainty about the fraction of total possible orders that are placed. This implies that on average, buyers and non-buyers will have similar beliefs about the aggregate order.

Our main theorem shows that the market order mechanisms will be competitive in the large if the price impact and the span both vanish in the limit.

Importantly, while market order mechanisms can guarantee competitive outcomes, they generally do not aggregate information, in the sense that the market price need not converge in probability to the marginal buyer's expected value. It is natural to ask whether there are other trading protocols, perhaps outside the classes of limit order and market order mechanisms, that guarantee information aggregation in an analogous manner. In Section 4, after proving our main theorem, we offer a number of examples that demonstrate the impossibility of guaranteeing information aggregation in any reasonable sense. Simply put, this is because there are information structures for which no mechanism can get close to aggregating the buyers' information in any equilibrium, no matter how large the market is. Indeed, as we suggested previously, the fact that market order mechanisms do not try to aggregate information may be seen as a feature rather than a bug: In order to aggregate information, it would be necessary to give buyers with large amounts of information significant influence over market prices. But granting such control would be self-defeating: informed buyers would use their influence to manipulate the market price, and push it below the marginal value. By limiting buyers' influence on prices, market order mechanisms with a small price impact limit the scope for price manipulation. This is a key aspect of how they guarantee competitive outcomes in large markets.

How should we interpret these results? One possibility is that the goods belong to a monopolistic seller who can commit to a trading protocol. Our theorem implies that such a seller could use market order mechanisms with well-chosen pricing rules to extract as much surplus as if no one had any private information about the common component. Moreover,

given the weak assumptions of our theorem, this level of revenue is unimprovable, and is the best that one can hope to guarantee, given that buyers have private information about their idiosyncratic components of values.³ Specific markets which we think fit our model reasonably well include agricultural commodities, natural resources, common stock sold via IPOs, government debt, or event tickets.

Another interpretation of models of strategic trading in large markets, common throughout the literature, is that the mechanism serves as a metaphor for decentralized market structures. This is an alternative to a limit model in which buyers are price takers, and the solution concept is rational expectations equilibrium, wherein the market price depends directly on fundamentals, which are in turn revealed through aggregate behavior of buyers. As pointed out long ago by Beja (1977), such models can admit paradoxical solutions in which prices are supposed to incorporate buyers' private information about fundamentals, but then buyers may ignore their information and trade just based on the price.⁴ In the large but finite strategic models, such paradoxes are precluded by explicit assumptions about the timing of how buyers' actions determine prices, and information aggregation in the large market limit is viewed as a foundation for efficient rational expectations equilibrium.

Following this tradition, we also explore the market order mechanism as metaphor for decentralized trade. To be clear, our objective is not to provide a foundation for efficient rational expectations equilibrium. Rather, we take seriously the market order mechanism as a metaphor for decentralized trading in large markets, and explore its implications for market outcomes.

An important distinction between market order and limit order mechanisms concerns how buyers may implicitly use the information contained in prices. In particular, in a limit order mechanism, a buyer's action is a constraint on the prices at which they may trade. This feature of the mechanism is loosely aligned with the idea that buyers can refuse to trade after they see the price. By contrast, in a market order mechanism, if a buyer agrees to trade, then they are bound to accept the market price if their order is fulfilled, whatever that price may be.

Our view is that both of these classes of mechanisms are crude static approximations to what is really a complex dynamic process by which buyers determine how to trade. The

³In particular, one possibility is that the idiosyncratic components are independent across buyers and perfectly correlated across units for a single buyer. In that case, revenue generation in the large market reduces to a standard monopoly pricing problem, with the optimal mechanism being to randomize between at most two posted prices. This randomization could be achieved by splitting the market into two groups, with two different quantities of the good. Within each group, market order mechanisms would extract the optimal revenue. We discuss this issue further in Section 3.

⁴Indeed, it is a logical impossibility that each buyer faces a price that incorporates all of the buyers' private information, and yet the buyers only act after seeing prices.

extent to which trades are actually conditioned on the information contained in prices lies somewhere in between the two extremes of full conditioning and no conditioning. Certainly, the natural assumption is that at the point of sale, a buyer sees the price being offered and then decides whether or not to purchase. But that final purchase decision may be preceded by a series of choices that affect the proximate willingness to pay for the good once prices are revealed. For example, when pulling up to the gas station with an empty tank, a driver could decide not to purchase fuel after seeing the posted price per gallon. But not purchasing fuel would be tantamount to abandoning the vehicle. A more reasonable interpretation is that the decision to purchase gasoline was effectively made much earlier, when the driver chose to go on the trip that depleted the tank.⁵ If we view the mechanism as a static metaphor this kind of richer dynamic decision process, it may be reasonable to model buyers as choosing to purchase with rational expectations about prices but before they know the realized price. Of course, while market order mechanisms require buyers to commit to trades before prices are realized, nothing in our model rules out the possibility that buyers' information is sufficient for them to predict what the aggregate order, and hence the price, will ultimately be.

Section 5 of the paper explores this idea more fully. We interpret the pricing rule as an aggregate supply curve that is induced by competition between producers who have no private information. If it so happens that the price impact and span are both negligible, then the equilibrium outcome is close to efficient in all information structures and equilibria. A small span corresponds to a supply curve that is close to being perfectly inelastic; the good can be supplied at negligible cost up to a threshold, and beyond that point, the production cost jumps up to a level at which it is always inefficient to produce. This is a demanding condition for sure, and one that will not hold in many markets of interest. However, we also show that as long as price impact is negligible, then even though the outcome may be inefficient, welfare can never fall below what is obtained if the traders have no information at all. In that sense, some information is welfare enhancing relative to no information.

1.3 Related Literature

We conclude this introduction with a discussion of related literature. We have already noted the large body of work studying competitive behavior in large markets with private information. Wilson (1977) considered the case of a single unit for sale via a first-price auction, with pure common values, where buyers' signals are conditionally independent given the value (the so-called "mineral rights" model). Milgrom (1979) generalized the analysis

⁵Of course, a high gas price may induce the driver to purchase only a partial tank, so that some adjustment is possible after seeing prices. This is consistent with the view just expressed that trades can be partially conditioned on the information contained in prices.

with conditionally independent signals, where the distribution could change with the number of bidders, and provides necessary and sufficient conditions for information aggregation. Milgrom (1981) further generalizes the analysis to uniform price auctions with single unit demand and a fixed number of units as the market grows large. Pesendorfer and Swinkels (1997, 2000) also study uniform price auctions with single unit demand, but where the number of available objects grows with the size of the market, as in our model. Pesendorfer and Swinkels (1997) show that information is aggregated the unique symmetric equilibrium when there are pure common values and conditionally independent signals that are ordered by monotone likelihood ratio. Pesendorfer and Swinkels (2000) add idiosyncratic components to values, again as in our model, and show that in equilibrium, information will be aggregated in that the price converges to the value of the marginal buyer. Jackson and Kremer (2007) study discriminatory auctions in the mineral rights model and show that the outcome is not competitive when the market grows large. Kremer (2002) distinguishes between information aggregation and competitiveness (which he refers to as “revenue efficiency”) in a pure common value environment and shows that it is possible to induce competitive outcomes without information aggregation.

Our model nests all of these as special cases, and our theorem shows that even when uniform price auctions will not aggregate information, market order mechanisms will still be competitive in the large. This is a rather striking result, given the weaknesses of our assumptions. Indeed, all of the literature just reviewed assumes very particular parametric forms for information, namely that signals are conditionally independent given the common component. Moreover, signals are linearly ordered and conditional distributions are stochastically ordered in such a way that there is an equilibrium in monotone pure strategies. A notable exception to that last statement is Pesendorfer and Swinkels (2000), who have two-dimensional private information: An idiosyncratic component and a one-dimensional, conditionally independent signal about the common component. Pesendorfer and Swinkels (2000) give conditions under which any sequence of equilibria must aggregate information, but they do not prove that equilibria exist. In fact, Jackson (2009) has shown that equilibria sometimes do not exist, in which case the conclusion about information aggregation is vacuous. By contrast, we make no assumptions at all about the structure of the buyers’ information: The idiosyncratic component associated to each unit is real-valued and marginal value is assumed to be increasing in it. But in terms of information, the buyers’ idiosyncratic types, the common component, and any additional signals the buyers observe may be correlated in an arbitrary manner. Moreover, no assumption is made about the structure of the equilibrium, but since there are finitely many buyers, types, signals, and actions, equilibria are guaranteed to exist, so that our theorems are not vacuous. This demonstrates the power of our proof

technique, which utilizes a lower bound on revenue obtained via weak duality, and therefore does not depend on a characterization of the equilibrium under any particular information structure.

Lauermann and Wolinsky (2017) and Lauermann and Speit (2023) consider first-price auctions in which the number of buyers is correlated with values, which introduces a novel form of adverse selection that may lead to non-competitive outcomes even in large markets. In an earlier version of this paper, we showed how the market order mechanisms guarantee competitive outcomes in large markets even when the number of buyers may be correlated with values.

The literature just reviewed concerns one-sided markets. Reny and Perry (2006) consider a two-sided market involving buyers and sellers who interact via a uniform price double auction. Payoffs depend on common and idiosyncratic components, with the idiosyncratic components functioning as signals for the common component and being conditionally independent given the common component. They prove that for generic value functions and for sufficiently large markets, there exist monotone equilibria, and that the equilibrium price converges to the price that clears the market ex post. Barelli, Govindan, and Wilson (2024) extend this analysis and give examples where no mechanism with a uniform price can aggregate information in large two-sided markets. In Section 4, we also give examples where it is impossible to aggregate information with uniform prices, but we also show the impossibility of information aggregation with arbitrary mechanisms. Relatedly, Du and Zhu (2017) construct a linear equilibrium in a multi-unit, uniform-price double auction with common and idiosyncratic components in value following normal distributions; they show that even asymptotically, the dispersed information about the common component is not fully aggregated in the equilibrium price due to the confounding effect of the idiosyncratic components. Throughout our paper, we consider one-sided market order mechanisms. Extending the analysis of market order mechanisms to settings with two-sided private information is a natural and important direction for future research.

Our paper is not the first to study market order mechanisms. Indeed, the celebrated model of Kyle (1985) is one in which a single, informed strategic trader and a non-strategic fringe of noise traders place market orders. A market maker then sets a price to clear the market. Lambert, Ostrovsky, and Panov (2018) extend a static version of the Kyle (1985) model to multiple strategic traders with asymmetric information about the common value. Ostrovsky (2012) introduces general information structure to the dynamic version of the Kyle (1985) model, in which traders trade with different prices at different times, and exogenous noise traders generate demand and supply over time as well. He characterizes when information is eventually aggregated, although along the way traders do not transact in prices that are

close to values. An important difference in our model is that we assume that each trader’s possible maximum order is small relative to the overall size of the market. Indeed, one could view the maximum trade as a design choice, so as to limit the amount by which any given buyer can manipulate prices and market outcomes.⁶ Further differences are that we focus on an one-sided market without noise traders and allow for idiosyncratic components in value.

This paper also contributes to the literature on informational robustness in mechanism design. Du (2018) and Brooks and Du (2021) study large markets where there is a single unit for sale and pure common values. Those papers construct sequences of auctions for which expected revenue converges to the expected value in all equilibria under all information structures. By contrast, the present paper studies a much richer model where there are many units for sale, multi-unit demand, and both common and idiosyncratic components to values. We also assume that buyers know at least their idiosyncratic component, which effectively imposes a lower bound on the buyers’ information. While in this paper we assume that the number of units grows proportionally with the population, in an earlier working paper, we demonstrated that analogous results hold if the number of units is fixed or grows at a strictly slower rate than the population. In the case of a single unit and pure common value, that version of the model nests that of Du (2018) and Brooks and Du (2021), and shows that asymptotic full surplus extraction can be achieved with market order mechanisms in which each buyer either places an order for one unit or zero units. Such mechanisms are much simpler than both exponential price mechanisms of Du (2018) or the proportional auctions of Brooks and Du (2021), each of which features a continuum of actions for the buyers.

On a methodological level, our proof technique involves computing a lower bound on expected revenue or expected welfare that holds across all Bayes correlated equilibria (BCE) (Bergemann and Morris, 2016) of the market order mechanism. A similar approach is taken in Du (2018) and Brooks and Du (2021, 2024). However, beyond that high level similarity, there are significant differences in the structure of the argument. In particular, in the preceding papers, there was no lower bound on the buyers’ information, so that it is possible to take the equilibrium action as an implicit description of everything a buyer knows. The preceding papers also considered mechanisms in which actions are ordered and the critical equilibrium constraints are local and pointing in the same direction, away from a non-participation action. By contrast, in the present paper, each buyer’s behavior is conditioned on both their equilibrium action and their idiosyncratic type. Moreover, with market order mechanisms, the conclusion that the outcome will be competitive in the large relies on equilibrium constraints in both “directions” of buying one more or one fewer unit, conditional on the idiosyncratic type profile. Thus, our results do not follow from any of the results of Du (2018), Brooks and

⁶We thank Piotr Dworczak for suggesting this interpretation of the assumption.

Du (2021, 2024) and related papers, or an immediate extension thereof. See also Footnote 9 for a more detailed discussion.

2 Model

2.1 Fundamentals

There are K units of a good available for sale. There are also N buyers, each of whom demands M units of the good.

Buyer i has a value of $v(\nu, \omega_{i,1}) \in [0, \bar{v}]$ for the first unit of the good, $v(\nu, \omega_{i,2})$ for the second unit, and more generally $v(\nu, \omega_{i,m})$ for the m -th unit, $1 \leq m \leq M$, where $\nu \in \mathcal{V}$ is the common component in the buyers' values, and $\omega_{i,m} \in \Omega \subseteq [\underline{\omega}, \bar{\omega}]$ is buyer i 's idiosyncratic component in the value for the m -th unit. We also refer to $\omega_i = (\omega_{i,m})_{1 \leq m \leq M}$ as buyer i 's idiosyncratic type (profile). When $M = 1$, we abuse the notation by writing $\omega_i \equiv \omega_{i,1}$. As an example, ν could represent the resale value of the good, while $\omega_{i,m}$ is a private use value.

Suppose $\mathcal{V}$ is a compact set, $v : \mathcal{V} \times [\underline{\omega}, \bar{\omega}] \rightarrow [0, \bar{v}]$ is continuous, and $v(\nu, \omega_{i,m})$ is weakly increasing and continuously differentiable in $\omega_{i,m}$. A leading example is the additively separable form $v(\nu, \omega_{i,m}) = \nu + \omega_{i,m}$. When $v(\nu, \omega_{i,m})$ is constant in $\omega_{i,m}$, we have the special case of pure common value.

The joint distribution of the common and idiosyncratic components is denoted by $\mu \in \Delta(\mathcal{V} \times \Omega^{NM})$. We assume μ places probability 1 on $\omega_{i,1} \geq \omega_{i,2} \geq \dots \geq \omega_{i,M}$ for every i , so the marginal value for additional units is weakly decreasing.

2.2 Information

The buyers know their idiosyncratic type profile $\omega_i = (\omega_{i,1}, \dots, \omega_{i,M})$. Their private information about the common component ν and others' idiosyncratic types is described by an *information structure* $I = (\prod_{i=1}^N (\Omega^M \times S_i), \sigma)$: each $(\omega_i, s_i) \in \Omega^M \times S_i$ is buyer i 's signal (or type), and $\sigma \in \Delta(\mathcal{V} \times \Omega^{NM} \times S)$ is the joint distribution of values and signals, where $S = \prod_{i=1}^N S_i$. We assume that the sets S_i and Ω are finite, though they can be arbitrarily large. We let $\mathcal{I}(\mu)$ be the set of all information structures $(\prod_{i=1}^N (\Omega^M \times S_i), \sigma)$ where $\text{marg}_{\mathcal{V} \times \Omega^{NM}} \sigma = \mu$.

Note that we allow for arbitrary correlation between ω_i , s_i and ν . For example, a buyer with a high ω_i may receive a pessimistic signal s_i about ν , so that types may not be linearly ordered in terms of their first order beliefs about values, or in terms of their higher order beliefs for that matter.

2.3 Market Order Mechanisms

Trade occurs through a *market order mechanism*: Each buyer takes an action $a_i \in A_i = \{0, 1, \dots, M\}$, interpreted as an order to buy a_i units. Given an action profile $a \in A = \{0, 1, \dots, M\}^N$, the buyers are randomly ordered and their orders are fulfilled one by one according to the order until the supply is exhausted. That is, let $\iota : \{1, 2, \dots, N\} \rightarrow \{1, 2, \dots, N\}$ be a bijection (uniformly drawn from the set of all bijections), and let $\bar{k}$ be the largest integer such that $\sum_{k=1}^{\bar{k}} a_{\iota(k)} \leq K$; then for each $k \leq \bar{k}$, buyer $\iota(k)$ receives $a_{\iota(k)}$ units of the good, and for each $k > \bar{k}$, buyer $\iota(k)$ does not receive any units. Let $q_i(a)$ be the ex ante probability that the a_i units of buyer i is fulfilled, in expectation across all orders ι . Clearly, when $M = 1$, we have $q_i(a) = r(\Sigma a)$, where $\Sigma a \equiv \sum_i a_i$, and

$$r(n) \equiv \min\{K/n, 1\}. \quad (1)$$

Moreover, $q_i(a)$ is close to $r(\Sigma a)$ when N and K are large and M is fixed (see Lemma 2 in Section A.1).

In addition, buyers pay a uniform price of $p(\Sigma a)$ per unit, where $p : \{0, 1, \dots, NM\} \rightarrow [0, \mathcal{C}]$ is a pricing rule, and $\mathcal{C}$ is a constant that is larger than $\bar{v}$.

2.4 Equilibrium

Given an information structure I , a (behavioral) *strategy* for buyer i in the market order mechanism is a mapping $b_i : \Omega^M \times S_i \rightarrow \Delta(A_i)$. Buyer i 's expected utility given a strategy profile b and the pricing rule p is

$$U_i(p, I, b) \equiv \int_{\nu, \omega, s, a} q_i(a) \sum_{m=1}^{a_i} (v(\nu, \omega_{i,m}) - p(\Sigma a)) \left(\prod_j b_j(a_j | \omega_j, s_j) \right) \sigma(d\nu, d\omega, ds), \quad (2)$$

A (*Bayes-Nash*) *equilibrium* for the game (p, I) is a strategy profile b such that for every buyers i and for every strategy b'_i , $U_i(p, I, b) \geq U_i(p, I, (b'_i, b_{-i}))$. Let $\mathcal{E}(p, I)$ be the set of equilibria for (p, I) . Because actions and signals are finite, the set of equilibria is always non-empty.

Let $R(p, I, b)$ be the expected revenue from at an equilibrium b :

$$R(p, I, b) \equiv \int_{\nu, \omega, s, a} \sum_i a_i q_i(a) p(\Sigma a) \left(\prod_j b_j(a_j | \omega_j, s_j) \right) \sigma(d\nu, d\omega, ds).$$

Define the *revenue guarantee* of the market order mechanism with pricing rule p under the prior μ as the infimum expected revenue over all information structures $I \in \mathcal{I}(\mu)$ and all equilibria $b \in \mathcal{E}(p, I)$:

$$\underline{R}(p, \mu) \equiv \inf_{I \in \mathcal{I}(\mu)} \inf_{b \in \mathcal{E}(p, I)} R(p, I, b).$$

The revenue guarantee is also the minimum expected revenue across all *Bayes correlated equilibria* (Bergemann and Morris, 2016, see their Definition 2 and the discussion therein).

2.5 Sequences of Economies

We will consider sequences of economies, where N goes to infinity. Along these sequences, we hold the value function v , the price upper bound $\mathcal{C}$ and the units of demand M as fixed, but the rest of the economy is allowed to vary. In particular, there are sequences of numbers of units (K_N), joint distributions of common and idiosyncratic components (μ_N), information structures (I_N), and pricing rules (p_N).

We assume that there is a constant $\kappa \in (0, M]$ such that K_N/N converges to κ as N goes to infinity. Thus, the number of units is assumed to grow proportionally with the population.

We make the following assumption about μ_N as $N \rightarrow \infty$. Let $F_N(\cdot; \omega)$ be the empirical cumulative distribution function (CDF) for ω , i.e.,

$$F_N(z; \omega) = \frac{|\{(i, m) : \omega_{i,m} \leq z\}|}{NM}$$

for $z \in \mathbb{R}$. There exists a (deterministic) CDF F to which the empirical CDF F_N converges in probability as $N \rightarrow \infty$. That is, for every $\delta > 0$,

$$\lim_{N \rightarrow \infty} \mu_N \left(\sup_{z \in [\omega, \bar{\omega}]} |F_N(z; \omega) - F(z)| > \delta \right) = 0. \quad (3)$$

Moreover, we suppose F is absolutely continuous and strictly increasing on $[\omega, \bar{\omega}]$.

Condition (3) is an assumption on μ_N 's marginal distribution of idiosyncratic types ω and does not restrict at all its marginal on the common component ν , or the correlation between ω with ν or across or within buyers. The condition says that uncertainties about the empirical distribution of idiosyncratic components must vanish as the market gets large. Since each buyer observes their ω_i , they know with very high precision their quantiles in the empirical distribution for each of their M marginal values. For example, if in μ_N the idiosyncratic components are N independent draws from a $\rho \in \Delta(\Omega^M)$, then by the Glivenko-Cantelli

theorem condition (3) is satisfied for

$$F(z) = \frac{1}{M} \sum_{m=1}^M \sum_{x \in \Omega: x \leq z} \rho_m(x),$$

where ρ_m is the marginal of ρ on the m -th component.

Let F^{-1} be the quantile function: $F^{-1}(r) = \inf\{z \in \mathbb{R} : F(z) \geq r\}$ for $r \in [0, 1]$. Thus, $F^{-1}(r)$ is the r -th percentile idiosyncratic component, and because F is continuous and strictly increasing, $F(F^{-1}(r)) = r$.

Fix a parameter $x \in [0, \kappa/M]$. A sequence of market order mechanisms with pricing rules (p_N) is *guaranteed to be competitive in the large (at x)* if for every sequence of priors (μ_N) satisfying (3),

$$\lim_{N \rightarrow \infty} \left(\frac{R(p_N, \mu_N)}{NMx} - \int_{\nu, \omega} v(\nu, F^{-1}(1-x)) \mu_N(d\nu, d\omega) \right) = 0. \quad (4)$$

In words, the sequence is competitive in the large if, in the limit as N goes to infinity, the market order mechanism is guaranteed to sell approximately NMx units at an expected equilibrium price equal to the expected value of the marginal buyer. This expected marginal buyer's value is the price that would clear the market, if no buyer had any private information about ν . The parameter x controls the fraction of the maximum order that is filled for the average buyer. If we take $x = \kappa/M$, then the definition would require that almost all the units sell out.

3 Competition in the Large via Market Order Mechanisms

We now present our main result, which consists of a pair of conditions such that market order mechanisms are guaranteed to be competitive in the large.

The *price impact* of a pricing rule p is $\gamma = \max_n |p(n+1) - p(n)|$. Note that γ may vary with N , as p depends on N .

Fix a $x \in (0, 1)$. We say that $\underline{p}$ is an *admissible low price* for x if either $\underline{p} < \min_{\nu} v(\nu, F^{-1}(1-x))$, or $\underline{p} = \min_{\nu} v(\nu, F^{-1}(1-x)) = 0$. The pricing rule p *spans* $(\underline{p}, x, \epsilon)$ if (i) $\underline{p}$ is an admissible low price for x , (ii) $p(n) \leq \underline{p}$ for $n \leq NM(x - \epsilon)$, and (iii) $p(n) \geq \bar{v}$ for $n \geq NM(x + \epsilon)$. Intuitively, the span represents the range of aggregate orders where the price transitions from sufficiently low (guaranteeing the goods sell out) to prohibitively high (choking off demand).

Fix a sequence of pricing rules (p_N) with associated price impacts γ_N . We say that the sequence has *vanishing price impact* if $\gamma_N \rightarrow 0$. We say that the sequence has *vanishing span at x* if there is a sequence (ϵ_N) where p_N spans $(\underline{p}, x, \epsilon_N)$, and $\epsilon_N \rightarrow 0$.

As an example, consider the following piecewise-linear pricing rules:

$$p_\epsilon(n) = \begin{cases} \underline{p} & \text{if } n \leq NM(x - \epsilon); \\ \underline{p} + (\bar{v} - \underline{p}) \frac{n - NM(x - \epsilon)}{2NM\epsilon} & \text{if } NM(x - \epsilon) < n \leq NM(x + \epsilon); \\ \bar{v} & \text{if } n > NM(x + \epsilon), \end{cases} \quad (5)$$

Suppose $\underline{p}$ is an admissible low price for x . If $N\epsilon_N \rightarrow \infty$, then the sequence (p_{ϵ_N}) has vanishing price impact. If $\epsilon_N \rightarrow 0$, then the sequence (p_{ϵ_N}) has vanishing span at x . Finally, if both $\epsilon_N \rightarrow 0$ and $N\epsilon_N \rightarrow \infty$ (say if $\epsilon_N = 1/\sqrt{N}$) then the sequence has both vanishing price impact and vanishing span at x . More broadly, for price impact to vanish, it is necessary that the span not vanish too quickly, and vice versa.

Theorem 1. *Suppose (p_N) is a sequence of pricing rules that has vanishing price impact and vanishing span at $x \in (0, \kappa/M]$. Then the associated sequence of market order mechanisms is guaranteed to be competitive in the large at x .*

As we said in the introduction, an immediate implication of Theorem 1 is that a monopoly seller could use market order mechanisms to sell whatever fraction of the goods they wish at a price that is equal to the expected marginal value. One might ask whether or not a seller could do even better, perhaps with other mechanisms. In fact, under our weak assumptions about the joint distribution of common and idiosyncratic components, the revenue guarantee provided by market order mechanisms is unimprovable. This is easiest to see in the case of pure common values, for which the limiting revenue guarantee is just the expected value times the number of units sold, which when $x = \kappa/M$, is equal to the efficient surplus. Even when there are idiosyncratic components to values, one possibility is that the ω_i are independent across i and independent of ν , and with probability one $\omega_{i,1} = \dots = \omega_{i,M}$, meaning that within a buyer, the idiosyncratic component is constant across units. When buyers have no additional information beyond ω_i , there are effectively independent private values, where the inverse demand when selling NMx units is

$$P(x) \equiv \int_{\nu, \omega} v(\nu, F^{-1}(1 - x)) \mu_N(d\nu, d\omega).$$

When the associated revenue function $NMxP(x)$ is concave in x (the regular case of Myerson (1981)), maximizing over $x \in [0, \kappa/M]$ gives the asymptotically optimal revenue in (4). Even

if the revenue curve is not concave, optimal revenue in the limit would either be obtained in the same manner via a deterministic sale of NMx units, or by randomizing over at most two quantities x and x' with probabilities α and $1-\alpha$ respectively, such that $\alpha x + (1-\alpha)x' = \kappa/M$. In this case, optimal revenue could be achieved in the limit as the market grows large by randomly splitting buyers into two groups, with probability α of being sorted into the first group, and using a market order mechanisms to sell αK units to the first group and $(1-\alpha)K$ units to the second group. The bottom line is that, aside from superficial randomization, market order mechanisms are sufficient for maximizing the seller’s revenue guarantee in the limit as the market grows large.⁷

In an earlier version of this paper, we also characterized the rate of convergence of the revenue guarantee per unit in the case of pure common values, which turns out to be the maximum of the rates at which the price impact and the span vanish. The maximum is minimized when both converge to zero at the same rate, which is equal to $1/\sqrt{N}$. Brooks and Du (2021) show that this is the optimal rate when there is only one unit available ($K = 1$), but we do not know whether this rate can be improved when there are multiple units for sale ($K > 1$). When there are also idiosyncratic components to values, the rate of converge of the revenue guarantee per unit also depends on the rate at which the empirical distribution of ω converges to its limit F .

3.1 Proof sketch for Theorem 1

We now sketch the argument for Theorem 1; a detailed proof is found in Sections A.1 and A.2.

We first reduce the analysis to the unit demand case. In particular, an equivalent way to represent the market order mechanism is the following: each buyer i has M “accounts” numbered 1 through M ; through each account buyer i can submit a market order for one unit of the good; the unit acquired by account m generates a value of $v(\nu, \omega_{i,m})$ to buyer i . The price $p(n)$ is determined by the total number n of market orders submitted by all accounts of

⁷This is the highest guarantee one could hope to achieve, based only on the knowledge that the empirical distribution of ω converges to F . With additional information about μ_N , the seller might be able to do even better, by leveraging the correlation in ω and applying Crémer and McLean (1988). Moreover, the seller might know that buyers with lower indices happen to have lower ω ’s, so he should charge them lower prices. Or it could be that each individual’s demand is known to be strictly downward sloping, in which case restricting the set of possible orders (effectively bundling units together) might lead to higher revenue. Furthermore, the seller could in general do better by having the quantity sold depend on the realized empirical distribution of $(v(\nu, \omega_{i,m}))_{i,m}$.

all buyers, and the probability that all of buyer i 's orders are fulfilled is determined by the random order over the buyers as in Section 2.⁸

We will prove that, up to terms that are lower order than N , expected revenue of the market order mechanism in the original economy is bounded below by expected revenue in an alternative “separated” economy, where each account belongs to a different buyer with unit demand (for a total of NM buyers). There are two parts to this argument: First, we obtain a lower bound on what happens in the original economy by considering just those equilibrium constraints that correspond to deviating in one account at a time, which are still feasible in the separated economy. With vanishing price impact, the gains from those deviations is close to what it would be if one ignored the effect of a deviation on other accounts. The second part is showing that the probability of being allocated a given unit m in the original economy, where the all-or-nothing rationing rule is used, is close to the probability of being allocated the same unit in the separated economy.

To be more precise, for the separated economy let (with a slight abuse of notation) $a_i = (a_{i,m})_{1 \leq m \leq M} \in \{0, 1\}^M$. Thus $a_{i,m} = 1$ if buyer i places a market order in account m , and 0 otherwise. Let us define the *strategic virtual objective* (SVO) for the market order mechanism in the separated economy as⁹

$$\lambda(\nu, \omega, a) = \min\{\Sigma a, K\}p(\Sigma a) + \sum_{i,m} \left[\alpha_{i,m}(\omega_{i,m}) \mathbb{I}_{a_{i,m}=0} (r(\Sigma a + 1)(v(\nu, \omega_{i,m}) - p(\Sigma a + 1))) \right. \\ \left. - \beta_{i,m}(\omega_{i,m}) \mathbb{I}_{a_{i,m}=1} (r(\Sigma a)(v(\nu, \omega_{i,m}) - p(\Sigma a))) \right], \quad (6)$$

where $\Sigma a = \sum_{i,m} a_{i,m}$, $\alpha_{i,m} : \Omega \rightarrow \mathbb{R}_+$, and $\beta_{i,m} : \Omega \rightarrow \mathbb{R}_+$. The SVO gives the following lower bound on the revenue guarantee in the original economy:

⁸Fix an information structure $I = (\Omega^{NM} \times S, \sigma)$ and an equilibrium b for the market order mechanism in Section 2, we can convert b to an equilibrium in the mechanism here under I by having buyer i placing orders on his accounts 1 to m with probability $b_i(m|\omega_i, s_i)$, for every signal (ω_i, s_i) and $m \in \{1, \dots, M\}$. Likewise we can convert any equilibrium in the mechanism here to one in the mechanism in Section 2 under any information structure.

⁹The term “strategic virtual objective” is also used in Brooks and Du (2021, 2024) to refer to the coefficient in a Lagrangian for guarantee maximization. As in our usage, that coefficient is equal to the designer’s payoff, plus multiplier-weighted sums of agents’ gains from deviation. There are however two important differences: First, in Brooks and Du (2021, 2024), the mechanisms being considered have actions that are ordered, with the lowest being a *secure action* that ensures agents are willing to participate in the mechanism, and the multipliers are non-zero only for deviations that are local and pointing away from the secure action. By contrast, in this SVO, the relevant deviations correspond to both increasing and decreasing the order. The second important difference is that in Brooks and Du (2021, 2024), the agents deviate based only on their equilibrium actions, whereas in our setup, each buyer also deviates on the basis of their idiosyncratic type profile. For those reasons, the analysis of the expected lowest SVO in the present paper proceeds along quite different lines than in prior work.

Proposition 1. *For any arbitrary $(\alpha_{i,m} : \Omega \rightarrow \mathbb{R}_+, \beta_{i,m} : \Omega \rightarrow \mathbb{R}_+)_{i,m}$, we have*

$$\begin{aligned} \underline{R}(p, \mu) \geq & \int_{\mu, \omega} \min_a \lambda(\nu, \omega, a) \mu(d\nu, d\omega) \\ & - \max_{i,m,\omega_{i,m}} (\alpha_{i,m}(\omega_{i,m}), \beta_{i,m}(\omega_{i,m})) NM(C/K + \gamma) \end{aligned}$$

for a constant C that only depends on $\bar{v}$, C and M .

Essentially, the SVO λ is a Lagrangian (see Lemma 1 in Section A.1) for minimizing expected revenue in the separated economy subject to the constraints that in equilibrium, conditional on their idiosyncratic type $(\omega_{i,m})$ and their action (either $a_{i,m} = 1$ or 0), the buyer must prefer that action to the alternative. These are referred to as obedience constraints, and $\alpha_{i,m}(\omega_{i,m})$ (respectively, $\beta_{i,m}(\omega_{i,m})$) is the Lagrange multiplier on the obedience constraint for the action $a_{i,m} = 0$ (respectively, $a_{i,m} = 1$). Proposition 1 implies that if these multipliers are chosen to be bounded as $N \rightarrow \infty$, and if $K_N \rightarrow \infty$ and $\gamma_N \rightarrow 0$, then the revenue guarantee per capita in the original economy is at least the expected value of $\min_a \lambda(\nu, \omega, a)/N$ when N is large.

Thus, we are led to analyzing $\min_a \lambda(\nu, \omega, a)$. Let us first illustrate the analysis for the case of pure common value ($v = \nu$). As a further simplification, suppose $p(n) = 0$ for $n \leq NM(x - \epsilon)$ and $p(n) = \bar{v}$ for $n \geq NM(x + \epsilon)$, so p spans $(0, x, \epsilon)$. In this case we set $\alpha_{i,m}(\omega_{i,m}) = A$ and $\beta_{i,m}(\omega_{i,m}) = 0$ for all i and m , where A is a constant. Then the SVO in (6) becomes

$$\lambda(\nu, \omega, a) = p(\Sigma a) \min(\Sigma a, K) + A(NM - \Sigma a)r(\Sigma a + 1)(v - p(\Sigma a + 1)),$$

and it is immediate that $\lambda(\nu, \omega, a)$ is decreasing in Σa when $\Sigma a < NM(x - \epsilon)$, and increasing in Σa when $\Sigma a \geq NM(x + \epsilon)$. When N is sufficiently large and ϵ is sufficiently small, for $\Sigma a \in [NM(x - \epsilon), NM(x + \epsilon)]$, $\min(\Sigma a, K) \approx NMx$ and $r(\Sigma a + 1) \approx 1$ since $K \approx N\kappa$ and $x \leq \kappa/M$, so we choose $A = \frac{x}{1-x}$, which implies the terms involving $p(\Sigma a)$ and $p(\Sigma a + 1)$ approximately cancel out and $\lambda(\nu, \omega, a) \approx xNMv$.

In the general case, we set $\alpha_{i,m}(\omega_{i,m}) = A$ and $\beta_{i,m}(\omega_{i,m}) = 0$ for $\omega_{i,m}$ above $F^{-1}(1 - x)$, and $\alpha_{i,m}(\omega_{i,m}) = 0$ and $\beta_{i,m}(\omega_{i,m}) = A$ for $\omega_{i,m}$ below $F^{-1}(1 - x)$. Subject to $\Sigma a = n$, we will show that the minimizing a for $\lambda(\nu, \omega, a)$ must have $a_{i,m} = 1$ whenever $\omega_{i,m}$ is among the top n highest idiosyncratic types. Moreover, as in the common value special case, the minimizing a must have Σa in the small neighborhood around xNM where $p(\Sigma a)$ is non-extreme, and A is chosen so that the terms in $\lambda(\nu, \omega, a)$ involving $p(\Sigma a)$ and $p(\Sigma + 1)$ approximately cancel out when N is large. In the end we get $\min_a \lambda(\nu, \omega, a) \approx xNMv(\nu, F^{-1}(1 - x))$.

In both cases we then apply Proposition 1 to conclude that the market order mechanisms are guaranteed to be competitive in the large at x .

3.2 On the Necessity of Vanishing Price Impact and Span

As a final task for this section, we show via two counterexamples that the price impact and span must both vanish in order for market order mechanisms to guarantee that the outcome is competitive in the large. Both examples feature pure common value ($v = \nu$) and unit demand ($M = 1$). Moreover, the common value v is either 0 or 1, with equal probabilities.

First, suppose $p_N(n) = c$ if $n \geq Nx$ and $p_N(n) = 0$ if $n < Nx$, where $x \in (0, 1)$ and $c \in (0, 1]$; so the pricing rules have non-vanishing price impact. When $c = 1$, regardless of the buyers' information there is always an equilibrium in which a buyer places an order if and only if $i < Nx$, so that revenue is exactly zero. When $c < 1$, suppose every buyer observes the realization of v . Then clearly the equilibrium is to play $a_i = 1$ when $v = 1$, and $a_i = 0$ when $v = 0$, leading to an expected revenue of $\kappa Nc/2 < \kappa N/2$. In both cases (p_N) fails to guarantee competitive outcomes regardless of how large N is. Clearly, the failure of competition stems from the large price impact: the action of the marginal buyer moves the price from 0 to c , and hence the price needs not to be close to the value of the marginal buyer.

Now suppose $p_N(n) = n/N$, so the pricing rules have non-vanishing span. Consider the following information structure: The signals are $S_i = \{0, 1\}$; if $v = 0$, then $s_i = 0$ for every buyer i ; if $v = 1$, then $s_i = 1$ with probability $\eta > 0$, and $s_i = 0$ with probability $1 - \eta$, independently across the buyers. Clearly, it is optimal to play $a_i = 1$ when $s_i = 1$. Let us look for a mixed-strategy equilibrium where conditional on $s_i = 0$, the buyer plays $a_i = 1$ with probability z . As $N \rightarrow \infty$, the indifference condition for such a buyer is

$$\frac{1 - \eta}{2}(1 - (\eta + z(1 - \eta))) + \frac{1}{2}(0 - z) = 0,$$

since by the law of large number the purchase rate (and hence, the price) is approximately $\eta + z(1 - \eta)$ when $v = 1$ and z when $v = 0$; let us assume $\kappa > \eta + z(1 - \eta)$ so there is never any rationing in the allocation. The above indifference condition solves to

$$z = \frac{(1 - \eta)^2}{1 + (1 - \eta)^2}.$$

As $N \rightarrow \infty$, the expected price in this equilibrium is approximately

$$\frac{1}{2}(\eta + z(1 - \eta)) + \frac{1}{2}z = \frac{1 - \eta}{2} + \frac{\eta}{2}(\eta + z(1 - \eta)) < \frac{1}{2}.$$

Thus, the expected equilibrium price is strictly less than the expected value even as $N \rightarrow \infty$. In particular, when $v = 1$, a large fraction of the buyers purchase the good at a price strictly below the value, earning a strictly positive rent. Intuitively, the large span of the pricing rule sustains a large difference between the expected values of buyers and the non-buyers; and in both states ($v = 0, 1$) the marginal buyer has the same pessimistic information ($s_i = 0$) as the non-buyers, causing a downward bias in the price.

4 Competition versus Information Aggregation

4.1 Information Aggregation

Our main result is that market order mechanisms are guaranteed to be competitive in the large. By contrast, much of the literature has focused on the question of whether or not information is aggregated. Information is aggregated if, roughly speaking, the price is equal to the marginal buyer's value. More formally, given a sequence of economies indexed by N with units K_N for sale, we will say that a sequence of mechanisms and equilibria *aggregates information* if for every $\epsilon > 0$, when the population N is sufficiently large,

- (i) The number of units allocated is at least $K_N(1 - \epsilon)$;
- (ii) With probability at least $1 - \epsilon$, at least a $1 - \epsilon$ fraction of the buyers who are allocated a unit pay a price per unit that is within ϵ of $\nu(v, F^{-1}(1 - K_N/(MN)))$.

Thus, the goods almost sell out, and almost all buyers pay a price close to the marginal valuation almost all the time. This is a weak condition in the sense that it only requires that most, but not all, buyers pay a price that is close to the marginal value. We note that while condition (i) has nothing directly to do with information, it is necessary to capture the idea that information is both aggregated and used for facilitating trade. Without (i), (ii) has no content.

Of course, if information is aggregated along some sequence of economies, mechanisms, and equilibria, then the limit is also competitive (at $x = K_N/(MN)$). But as pointed out by Kremer (2002), competitiveness does not imply information aggregation, because the former only requires the price per unit to be close to the marginal value $\nu(v, F^{-1}(1 - K_N/(MN)))$ in expectation.

Market order mechanisms induce competitive outcomes but they generally do not aggregate information. For example, suppose there is a pure common value, one buyer knows the value exactly, and the remaining $N - 1$ buyers have no information, as in Engelbrecht-Wiggans, Milgrom, and Weber (1983). Since that one informed buyer can only shift the aggregate

order by at most M units, then if the price impact goes to zero, the amount by which they can shift the price also declines to zero as the market grows large. It is therefore infeasible for all of the informed buyer's information to be incorporated into the price. However, there do exist other mechanisms and equilibria that would aggregate information for this sequence of information structures. Indeed, the mechanism could simply not offer to sell the good to the informed buyer under any circumstances, but still ask them what the value is, and then sell the goods to the remaining uninformed buyers at a price equal to the informed buyer's reported value. This mechanism aggregates information, since the goods sell out (or almost all of the goods if $K_N = NM$), and the price per unit is equal to the marginal buyer's value, which in this case is just the pure common value.

Beyond this example, it is well known that there are sequences of economies for which there do exist sequences of mechanisms and equilibria that aggregate information. However, there do not exist mechanisms that guarantee information aggregation for all sequences of information structures and equilibria, in the same way that market order mechanisms guarantee competition. We will now present four examples illustrating the main difficulties.

All of the examples feature single unit demand, a pure common value v that is either 0 or 1, and there are either $N/2$ or $(N + 1)/2$ units for sale. Because of the pure common value, if information is aggregated, then buyer surplus per capita must go to zero as N goes to infinity.

4.2 Impossibility with Uniform Prices

Market order mechanisms have the feature that all buyers pay the same price per unit, i.e., there is a uniform price. We now give an example in which it is impossible to guarantee information aggregation with a uniform price. Both values in $\{0, 1\}$ are equally likely. The buyers' information is as follows: If the value is 1, then exactly one buyer (chosen at random) observes a signal $s_i = 1$, and all other buyers observe $s_i = 0$. If the value is 0, then all buyers observe $s_i = 0$.

We claim that for these information structures, there is no sequence of mechanisms with uniform prices and associated equilibria in which the market clearing price converges in probability to the value. To obtain a contradiction, suppose that such a sequence did exist. Then for any $\epsilon > 0$, we can find N large so that when $v = 0$, at least $N(1/2 - \epsilon)$ units sell out and, moreover, at least a fraction $1 - \epsilon$ of those buyers pay at most ϵ when the value is zero. Thus, at least $N/4$ of the buyers must think that they get allocated a unit with ex ante probability at least $1/4$ when all buyers have signals $s_i = 0$.¹⁰

¹⁰If not, then $3N/4$ buyers get the good with probability at most $1/4$, and so the expected number of units sold is less than $N/4 + 3N/16 = 7N/16 < N(1/2 - \epsilon)$.

Consider the buyers who get the good with probability at least $1/4$ when the value is zero and who pay a price of at most ϵ per unit. There are at least $N(1 - \epsilon)/4$ of them. Now suppose that a buyer in this set observes $s_i = 1$. That buyer can deviate and behave as if they had seen $s_i = 0$, obtain the good with probability at least $1/4$, and pay a price of at most ϵ , for an expected surplus of at least $1/4(1 - \epsilon) > 1/8$ (for ϵ sufficiently small). This means that when such a buyer actually does observe $s_i = 1$, the expected equilibrium price must be less than $7/8$. But because there are $N/4$ buyers in this group, when $v = 1$, it must happen at least a quarter of the time that the price is less than $7/8$, so that information is not aggregated.

In this example, one buyer has an extraordinarily large amount of information, which can only be incorporated into the price by giving them a large amount of influence over the price. But because it is unknown who has the information *ex ante*, almost all buyers must be granted influence over the price. As a result, virtually all of the buyers have the ability to manipulate the mechanism and obtain a non-trivial amount of surplus.

4.3 Impossibility in all Equilibria

The second example illustrates the possibility that information can be aggregated along some sequence of equilibria, but for any sequence of mechanisms, there are also equilibria in which information is not aggregated. Now, buyers 1 and 2 each have a signal $s_i \in \{0, 1\}$, both values equally likely, and independently distributed. Other buyers have no information (so their signals are all zero). The value is equal to the parity of $s_1 + s_2$, i.e., the value is 0 if $s_1 + s_2$ is even and is 1 if the sum is odd.

Here are incentive compatible and individually rational direct mechanisms that aggregate information: All buyers are equally likely to be allocated the good, regardless of what they report (with probability close to $1/2$). If $s_1 + s_2$ is even, the price is 0, and if $s_1 + s_2$ is odd, the price is 1. This mechanism is incentive compatible and individually rational: Regardless of what a buyer reports, the expected allocation is the same, and the expected price is equal to the *ex ante* expected value.

Now, consider any N and any mechanism. We claim that there is an equilibrium where the strategies of buyers $i = 1, 2$ do not depend on s_i . Indeed, suppose that buyer 2's strategy does not depend on s_2 . Then the conditional distribution of (v, a_{-1}) is the same conditional on $s_1 = 0$ and on $s_1 = 1$, so that buyer 1 has a best response that is independent of s_1 . The same is true of buyer 2's best response when buyer 1's strategy does not depend on s_1 . This shows that the set of strategies that do not depend on signals is closed under best

responses, and therefore there is an equilibrium in strategies of that form. Clearly, if the buyers' strategies do not depend on s , then information is not aggregated.

With this example, the buyers have information, but each buyer has an incentive to use their information only if other buyers are using theirs as well. In such a situation, information may fail to be aggregated because the buyers coordinate on ignoring their own information. This is an example of what Ostrovsky (2012) refers to as a “non-separable” information structure. It is also a situation in which the null information structure in which the buyers have no information is a “coordinated individual garbling,” as defined by Brooks, Du, and Haberman (2025).

4.4 Impossibility in any Mechanisms and Equilibria

We now go a step further and exhibit information structures for which there is no sequence of mechanisms and equilibria that aggregate information. In this example, $v = 1$ with probability $2/3$ and $v = 0$ with probability $1/3$. Each buyer observes a signal s_i that is also either 0 or 1. Conditional on $v = 1$, we draw a number k uniformly from the odd numbers between 0 and N . Conditional on $v = 0$, we draw a number k uniformly from the *even* numbers between 0 and N . In either case, given the value of k that is drawn, exactly k buyers (chosen at random) observe $s_i = 1$, and the remaining $N - k$ observe $s_i = 0$.

Now, suppose that information is aggregated by some sequence of mechanisms and equilibria. Then the expected price conditional on v must be close to v with high probability. Let us write $q_i(v)$ for the probability that a buyer is allocated the good when the value is v , and let $t_i(v)$ be their expected payment. Then because of information aggregation, we have that

$$\frac{1}{N} \sum_i t_i(v) \approx \frac{v}{2}, \quad \frac{1}{N} \sum_i q_i(v) \approx \frac{1}{2}.$$

Now, suppose a buyer were to flip their strategy, and behave as if their signal were $1 - s_i$. Then the mechanism will necessarily behave the same as it would as if the value were $1 - v$, so the expected price will be close to $1 - v$, and the buyer's surplus is close to:

$$\frac{2}{3}(q_i(0) - t_i(0)) + \frac{1}{3}(-t_i(1)).$$

Since this is a lower bound on their surplus, we have that equilibrium buyer surplus per capita must be at least

$$\frac{1}{N} \sum_i \left[\frac{2}{3}(q_i(0) - t_i(0)) + \frac{1}{3}(-t_i(1)) \right] \approx \frac{1}{2} \left[\frac{2}{3}(1 - 0) + \frac{1}{3}(0 - 1) \right] = \frac{1}{6} > 0.$$

Thus, buyer surplus per-capita must be bounded away from zero, which contradicts information aggregation (which in the case of pure common values must drive buyer surplus per-capita to zero).

What is exceptional about this example is that no matter how large is the market, every buyer has a lot of information about the value. It is impossible to incentivize the revelation of that information without giving buyers a large fraction of the surplus. In comparison, the positive results for information aggregation in the literature either explicitly assume that the buyers are informationally small when the market is large (as in Bali and Jackson (2002)), or they posit information structures in which this is the case. Even though it is costly to elicit information when a significant fraction of buyers are informationally large, it is not hard to induce the buyers to compete away their rents using market order mechanisms.

This example is highly stylized, and we would not argue that it is a good reflection of buyers' information in any particular market. Nonetheless, it distills a fundamental tension that can preclude information aggregation: it is impossible to incorporate buyers' information into prices without also giving them control that they will be tempted to use to manipulate prices downwards.

4.5 Impossibility with Useful Information

Our last example concerns an issue which is, for the most part, outside of the literature. A premise that underlies the interest in information aggregation is that individuals will make decisions after seeing prices. Hence, information that is aggregated in prices can be incorporated into decision making, and therefore will facilitate more efficient allocation of resources in society. This is the essence of the efficient markets hypothesis. But the use of the aggregated information is largely outside the standard models of strategic trade in large markets. Indeed, even with limit order mechanisms, buyers choose their actions before they see prices. What if we insisted that buyers could make at least some decisions after seeing prices? How would that affect the market's ability to aggregate information?

To operationalize this idea, we will now suppose that after interacting in the mechanism, buyers see their assigned allocation and payments, and then may decide to back out and not purchase the good. This corresponds to "withdrawal rights" as in Haberman and Jagadeesan

(2025). What follows is an example in which it is possible to aggregate information with the standard interim participation constraint, but it is impossible to aggregate information if the buyers have withdrawal rights.

All of the buyers observe independent draws from $\{0, 1\}$ that are equally likely to be zero and one. The pure common value is $v = 0$ if $\sum_i s_i$ is even and $v = 1$ if the sum is odd. This is the same as the previous example, but now the value is equally likely to be 0 and 1 ex ante.

For any N , there is a mechanism and equilibrium that aggregate information: Have the buyers report their signals, allocate the goods at random, and set the price equal to the parity of the sum of the signals. Notice also that all buyers receive non-negative surplus conditional on their signal and conditional on the price.

However, for any mechanism and equilibrium that aggregates information, there is a way for a buyer to deviate and then exercise their withdrawal rights that is strictly profitable: First flip the strategy and behave as if the signal is $1 - s_i$. That will result in prices that, in expectation, are close to $1 - v$. For ϵ small, when the realized price is above ϵ , withdraw from trade, and when the price is less than ϵ , trade. This ensures that with high probability, the buyer's payoff will be close to $(1/2)q_i(0)(1 - \epsilon)$. For this to not be a profitable deviation, we must have that total buyer surplus must be nearly $N/4(1 - \epsilon)$, which contradicts information aggregation.

Our point is that it is not information aggregation per se that is desirable. It is that the aggregated information can be used to make more efficient decisions. Depending on how we model the way that strategic traders could use what they learn from prices, it may be that the ability to use the information contained in prices creates further opportunities for manipulation that leads information aggregation to unravel.

To be clear, we are not saying that markets or mechanisms should necessarily grant traders withdrawal rights. Market order mechanisms in particular do not give the buyers withdrawal rights after prices are made public. If buyers could withdraw after seeing the price in a market order mechanism, then there are examples in which they would want to do so.¹¹ As a result, market order mechanisms with withdrawal rights do not guarantee competition in the large. Nonetheless, we would again argue that, at least from a normative perspective, the lack of withdrawal rights can be viewed a feature rather than a bug: By requiring buyers

¹¹Indeed, consider the example from the beginning of this section, where only buyer 1 has any information and knows the value. Suppose for simplicity that the value is either 0 or 1, both equally likely. In a large market, there will be an equilibrium where just enough traders place orders so that the price is less than $1/2$ and one additional order pushes the price above $1/2$. Buyer 1 will place an order if and only if the value is 1. In that case, whether the realized price is above or below $1/2$ reveals the value, and the uninformed traders would withdraw if the price is below $1/2$.

to commit to trades before prices are realized, the mechanism shuts down a potential source of incentives to manipulate prices.

4.6 Discussion

Our point with these examples is not that information cannot be aggregated. Of course it can, under certain conditions and in certain equilibria. But this requires either that each buyer has relatively little information, or that relatively few buyers have a large amount of information and prices are discriminatory. It may also require limitations on the ways in which buyers can make use of the information they obtain through prices. In light of all of these complications, it is even more striking that market order mechanisms guarantee competitive behavior, regardless of the information structure and which equilibrium is played.

5 Decentralized Trade and Social Welfare

5.1 Prices and Purchases: Which comes first?

As we stated in the introduction, a leading application of our results is to the problem of a monopolistic seller who can commit to the rules by which they sell goods that are in their possession. Such a seller could choose to sell via a market order mechanism, with the pricing rule being an aspect of the design choice. As our theorem shows, by choosing a pricing rule with small price impact and span, such a mechanism operating in a large market would be assured to extract nearly all of the optimal revenue, no matter the information structure and equilibrium.

A rather different interpretation of our model is that there is no monopolistic seller or a designer with commitment power. Instead, the market order mechanism serves as a metaphor for a process of decentralized trade between buyers and sellers. In this interpretation, the pricing rule is not a designed object; it is an exogenous supply curve, induced by competition between producers. We now develop this interpretation of the model more fully, and then analyze range of possible welfare outcomes that could obtain.

Returning to a discussion in the introduction, market order mechanisms have the feature that buyers commit to purchase before knowing prices. On the face of it, this seems at odds with our common experience of how most trade actually takes place; we see prices and then choose whether or not to buy. So why would we suggest otherwise? We do not disagree at all that in most cases, the final purchase decision is made after seeing prices. But that may be just the final step in a complex and dynamic decision process. A static model of the market

aims to capture the entire process, and willingness to pay at the last step may not be the overall willingness to pay that is a meaningful reflection of the surplus generated by trade.

In the introduction, we illustrated this point with an example of a driver choosing whether or not to purchase gasoline. For another example, suppose your partner is going to the grocery store and they ask if you want anything. You give them a list of items. Of course, at the time you put it on the list, you have rational expectations about what each item will cost. But when your partner arrives at the store, the realized price may be higher or lower than what you imagined. Of course, they could call you if a price seems extraordinarily high. But to a first order, they will simply do what you ask, and purchase the items you requested at whatever are the realized prices. Clearly, the “value” in the model should belong to the person who put the item on the shopping list, not the value revealed by the behavior of the person doing the shopping.

Here is one last example. You invite someone out on a date. You skim menus and look at price ratings before making a reservation. The night comes, and you are out at the restaurant. Suppose that prices are higher than what you expected. In an extreme circumstance, you might leave and look for another restaurant on the fly. But a more reasonable scenario is that the difference between realized prices and your expectation at the time of making the reservation will be small compared to the harm from disrupting the date, and you would simply make the best of it. In that way, the true decision to purchase the meal was made in far advance, at the time of booking the reservation, and before prices were fully known. Moreover, the correct way to assign value to the meal is by revealed preference at the time of making the reservation, and not when deciding whether or not to abort after examining the full price list.

Of course, we picked examples in which purchase decisions are, to a large extent, made far in advance of the exchange of goods, and before price uncertainty is resolved, and for which market orders seem like a better metaphor than limit orders. We could also have given examples where limit orders seem like the better metaphor. The reality is somewhere in between the extreme benchmarks of trades fully conditioned on prices and trades unconditioned on prices. Even in the examples we have given, one can still at the margin adjust how much gasoline to buy, or which entree or bottle of wine to order. But situations in which buyers commit to purchases under significant price uncertainty seem to abound, e.g., acquiring a car that will require maintenance, starting a home renovation based on an initial estimate, or even moving to a new neighborhood or city. And while limit order mechanisms have been well explored as a metaphor for decentralized trade in large markets, the welfare properties of market order mechanisms in large markets appear to have been overlooked.

5.2 The Pricing Rule as a Supply Curve

Here is how we conceive of the pricing rule as a supply curve in a decentralized market. Suppose that there are $L \geq 2$ producers of the good. Seller l can supply k units of the good at cost $C_l(k)$. Let $c_l(k) = C_l(k) - C_l(k-1)$ be the marginal cost function of producer l . We assume that c_l is non-decreasing for all l . We assume the following sequential structure for how the market clears: First, each buyer i places their order $a_i \in \{0, \dots, M\}$. Then, after seeing the orders, each producer l posts a price p_l . The buyers who placed orders then choose from which producer to purchase. Finally, all orders are fulfilled, and the buyers pay the producer that they patronize. Note that we assume that producers can supply as many units as ordered, possibly at very high cost, so that we are also implicitly assuming that $K = NM$.

The subgame after the buyers have placed orders is a standard model of Bertrand competition. All equilibria of this model have the property that when n orders have been placed, the producers will compete the price down so that it is between the n th and $n+1$ th lowest marginal costs. In particular, we can define the aggregate cost function:

$$C(n) = \min\{C_1(n_1) + \dots + C_L(n_L) | n_1 + \dots + n_L = n\},$$

and the aggregate marginal cost $c(n) = C(n) - C(n-1)$. Then the equilibrium price is in the range $[c(n), c(n+1)]$, and the lowest marginal cost producers fill the orders. It does not matter especially which equilibrium is played in this subgame, but to fix ideas, we will suppose that the price is $p(n) = c(n)$, i.e., the marginal cost to produce the last unit.¹²

Given information I and strategies b , each buyer i 's utility is given by equation (2) with $q_i(a) = 1$; total welfare is

$$W(p, I, b) = \int_{\nu, \omega, s, a} \left(\sum_i \sum_{m \leq a_i} v(\nu, \omega_{i,m}) - \sum_{l \leq \Sigma a} p(l) \right) \prod_i b_i(a_i | s_i, \omega_i) \sigma(d\nu, d\omega, ds),$$

i.e., the total value of the units sold, less the production cost. As with revenue, we define the welfare guarantee $\underline{W}(p, \mu)$ as the minimum expected total welfare over all information structures with marginal μ and all equilibria:

$$\underline{W}(p, \mu) \equiv \inf_{I \in \mathcal{I}(\mu)} \inf_{b \in \mathcal{E}(p, I)} W(p, I, b).$$

¹²A hypothesis of Theorem 2 below is that price impact vanishes, in which case $|c(n) - c(n+1)| \rightarrow 0$.

If the buyers had no information except knowing their own idiosyncratic types, then welfare would be

$$W^*(p, \mu) = \max_n \int_{\nu, \omega} \sum_{l \leq n} (v(\nu, \omega^{(l)}) - p(l)) \mu(d\nu, d\omega), \quad (7)$$

where $\omega^{(l)}$ is the l -th highest value in $\omega = (\omega_{i,m})_{1 \leq i \leq N, 1 \leq m \leq M}$. In other words, the market would clear at a quantity where the supply curve intersects with the expected demand curve, where the expectation is taken over ν .

As in Section 3, we consider a sequence of priors (μ_N) that satisfies (3) for an absolutely continuous and strictly increasing CDF F , i.e., the uncertainty about the empirical distribution of idiosyncratic types vanishes as the market gets large.

5.3 An Inefficient Example

For any distribution μ_N , it is always possible that buyers know everyone's idiosyncratic type but have no information about the common component; if (μ_N) satisfies (3), then when N is large there is little uncertainty about the distribution of the idiosyncratic types in μ_N , so the equilibrium welfare would be close to $W^*(p_N, \mu_N)$. Thus,

$$\limsup_{N \rightarrow \infty} \frac{W(p_N, \mu_N) - W^*(p_N, \mu_N)}{N} \leq 0.$$

However, it is in general possible for this inequality to be strict, as the following example shows.

Suppose that there is a unit demand, a pure common value $v = 1/2$ (meaning there is no uncertainty about the value), and the pricing rule is $p(n) = (2/3)\mathbb{I}_{n > N/2}$. (We assume for this example that N is even.) So, the good is costless to produce to cover half the population, but above that point, the marginal cost jumps up to $2/3$. Under no information, the efficient outcome is for exactly half of the units to be sold, attaining a welfare of $N/4$, and buyers who purchase earn strictly positive surplus.

Now, consider the following information structure: Independently of the value, with probability $\eta \in [0, 1]$, exactly half of the buyers (chosen at random) receive a signal that tells them not to buy, and the other half receive a signal to buy. Otherwise, with probability $1 - \eta$, all of the buyers receive a signal telling them to buy. The strategies are such that the buyers follow their recommendations.

When $\eta = 1$, this is precisely the same outcome as under no information: half of the units are sold at a price of 0 in both states, and buyers who purchase get strictly positive rents.

Thus, for η close to 1, a buyer who purchases will still get a positive surplus in expectation. Moreover, a buyer who is told not to buy expects the price to be $1/2$, but if they were to purchase the price would jump up to $2/3$. Following the recommendations is therefore an equilibrium. But as η dips below 1, there is now a $1 - \eta$ probability event that an additional $N/2$ units are produced at a cost of $2/3$ per unit, even though the value of these additional units is only $1/2$. This is purely a loss to society from overproduction.

The reason that information can be harmful in this situation is that under no information, the marginal buyer earns strictly positive surplus. As a result, information can be used to coordinate the buyers on inefficient overconsumption that destroys surplus, without inducing them to withdraw their orders.

5.4 Welfare and Price Impact

The inefficiency in the preceding example relies on the fact that price impact is large. In fact, as the next result shows, when price impact is small, equilibrium welfare cannot be strictly below the no information benchmark in the large market limit:

Theorem 2. *Let (μ_N) be a sequence of priors satisfying (3), and let (p_N) be a sequence of supply functions with vanishing price impact. Then*

$$\lim_{N \rightarrow \infty} \frac{W(p_N, \mu_N) - W^*(p_N, \mu_N)}{N} = 0.$$

The proof of Theorem 2 is in the Appendix. It proceeds along very similar lines as the proof of Theorem 1. In particular, we construct an SVO for total welfare, and then compute a lower bound on the expected SVO that converges to the ex ante welfare benchmark. We first reduce the problem to the unit demand case, as in Section 3.1, and then we construct multipliers on the two equilibrium constraints of deviating from buying to not buying, or vice versa. The multipliers have an especially simple form: For buyers who would purchase in the no-information benchmark, there is a multiplier of 1 on deviating to buying when not buying, and for buyers who would not purchase under no-information, there is a multiplier of 1 on deviating to not buying when buying. All other multipliers are zero. We show that the resulting SVO is minimized at the aggregate order that solves W^* , which delivers the result.

Theorem 2 shows that in the large market, welfare per capita must be at least that under no information. A more demanding benchmark is the ex post efficient level of welfare, which we denote

$$W^{**}(p, \mu) = \int_{\nu, \omega} \max_n \sum_{l \leq n} (v(\nu, \omega^{(l)}) - p(l)) \mu(d\nu, d\omega).$$

We have the following corollary:

Corollary 1. *Let (μ_N) be a sequence of priors satisfying Condition (3), and let (p_N) be a sequence of pricing rules that has vanishing price impact and vanishing span at some $x \in (0, 1)$. Then we have*

$$\lim_{N \rightarrow \infty} \left(\frac{W(p_N, \mu_N)}{N} - \frac{W^{**}(p_N, \mu_N)}{N} \right) = 0.$$

As we mentioned previously, we regard the case with vanishing span as rather extraordinary; it implies that the efficient production level is independent of the common component in valuations, and hence $(W^{**}(p_N, \mu_N) - W^*(p_N, \mu_N))/N \rightarrow 0$ as $N \rightarrow \infty$. Vanishing price impact, on the other hand, seems like a more natural condition. But while information raises welfare relative to no information, the outcome may still be far from efficient ex post. This is an unavoidable possibility in a model where there is no lower bound on what the buyers may know about the common component. A natural and valuable direction for future research would be to assume that each buyer has some minimal information about the common component, while at the same time remaining agnostic about any additional information that the buyers may have and about what equilibrium is played, so that the buyers' information collectively reveals the common component in the large.

6 Conclusion

In this paper we have examined the revenue and welfare guarantees of market order mechanisms. A nice feature of market order mechanisms is their simplicity: the action is just an order for a number of units, compared with an entire demand curve in a limit order mechanism. As our proof shows, the remarkable welfare properties of market order mechanisms when the price impact and span are small depend only on the equilibrium constraints that buyers not want to purchase one more or one fewer unit. In particular, there is no need to consider other equilibrium constraints that correspond to larger changes in the order.

In the unit demand case, this has especially powerful implications for how buyers might learn the equilibrium: *any* no-regret learning dynamics would generate an average revenue/welfare over time in accordance to our revenue/welfare guarantees over informational structures and equilibria. Indeed, Hartline, Syrgkanis, and Tardos (2015) and Lomys and Magnolfi (2025) have shown that the average outcome over time from any no-regret learning dynamics is guaranteed to converge to the set of *Bayes coarse correlated equilibria* (the joint distribution over fundamentals and actions such that each player prefers their equilibrium strategy to any alternative strategy that always plays a fixed action conditional on their

idiosyncratic type; see Definition 3 of Lomys and Magnolfi (2025)); and when the action for each player is binary, Bayes coarse correlated equilibrium coincides with Bayes correlated equilibrium.

As a final remark on convergence to equilibrium, we note that because our bounds hold for all information structures and equilibria, there is absolutely no harm, and there may in fact be substantial gains, from facilitating communication between buyers in advance of running the mechanism and clearing the market. For example, the mechanism could allow trial rounds, in which buyers post provisional orders, and the mechanism provides feedback to buyers about what the resulting aggregate order and price would be. As long as at the last step the buyers are asked to submit final orders that will be binding, then our theorems apply. But pre-play communication might result in additional information for buyers that would reduce the set of possible equilibrium outcomes. For reasons entirely outside of our model, we suspect that such pre-play communication would be a powerful and beneficial element that would facilitate convergence of aggregate orders into the span.

A Omitted proofs

A.1 Proof of Proposition 1

As in Section 3.1, let $a_i = (a_{i,m})_{1 \leq m \leq M} \in \{0, 1\}^M$. Thus $a_{i,m} = 1$ if buyer i places a market order in account m , and 0 otherwise. Then buyer i 's ex post utility in the original economy is

$$u_i(\nu, \omega, a) = q_i(\Sigma a_1, \dots, \Sigma a_N) \sum_m a_{i,m} (v(\nu, \omega_{i,m}) - p(\Sigma a)),$$

where $\Sigma a = \sum_{i,m} a_{i,m}$ and $\Sigma a_i = \sum_m a_{i,m}$.

We first start with a standard Lagrangian relaxation result:

Lemma 1. *Fix arbitrary $\chi_i : \Omega^M \times \{0, 1\}^M \times \{0, 1\}^M \rightarrow \mathbb{R}_+$, $1 \leq i \leq N$. We have the following lower bound on the revenue guarantee:*

$$\underline{R}(p, \mu) \geq \int_{\nu, \omega} \min_{a \in \{0, 1\}^{NM}} \bar{\lambda}(\nu, \omega, a) \mu(d\nu, d\omega),$$

where

$$\bar{\lambda}(\nu, \omega, a) = p(\Sigma a) \min\{K, \Sigma a\} + \sum_i \sum_{a'_i} \chi_i(a_i, a'_i, \omega_i) (u_i(\nu, \omega, a'_i, a_{-i}) - u_i(\nu, \omega, a_i, a_{-i})). \quad (8)$$

Proof. Any equilibrium on an information structure induces an outcome $\sigma \in \Delta(\mathcal{V} \times \Omega \times A)$ whose marginal is μ ($\text{marg}_{\mathcal{V} \times \Omega^{NM}} \sigma = \mu$), and the obedience constraints hold for all i , ω_i , a_i and a'_i :

$$\sum_{a_{-i}, \omega_{-i}, \nu} (u_i(\nu, \omega_i, \omega_{-i}, a'_i, a_{-i}) - u_i(\nu, \omega_i, \omega_{-i}, a_i, a_{-i})) \sigma(\nu, \omega_i, \omega_{-i}, (a_i, a_{-i})) \leq 0.$$

Such an outcome σ is known as a *Bayes correlated equilibrium*.

Therefore, we have

$$\begin{aligned} & \sum_{a, \omega, \nu} \min\{K, \Sigma a\} p(\Sigma a) \sigma(\nu, \omega, a) \\ & \geq \sum_{a, \omega, \nu} \min\{K, \Sigma a\} p(\Sigma a) \sigma(\nu, \omega, a) \\ & \quad + \sum_{i, \omega_i, a_i, a'_i} \chi_i(a_i, a'_i, \omega_i) \left(\sum_{a_{-i}, \omega_{-i}, \nu} (u_i(\nu, \omega_i, \omega_{-i}, a'_i, a_{-i}) - u_i(\nu, \omega_i, \omega_{-i}, a_i, a_{-i})) \sigma(\nu, \omega_i, \omega_{-i}, (a_i, a_{-i})) \right) \\ & = \sum_{a, \omega, \nu} \bar{\lambda}(\nu, \omega, a) \sigma(\nu, \omega, a) \end{aligned}$$

$$\geq \sum_{\omega, \nu} \min_a \bar{\lambda}(\nu, \omega, a) \mu(\nu, \omega).$$

□

We set

$$\chi_i(\omega_i, a_i, a'_i) = \sum_m \left(\mathbb{I}_{a_{i,m}=0, a'_{i,m}=1, a_{i,-m}=a'_{i,-m}} \alpha_{i,m}(\omega_{i,m}) + \mathbb{I}_{a_{i,m}=1, a'_{i,m}=0, a_{i,-m}=a'_{i,-m}} \beta_{i,m}(\omega_{i,m}) \right), \quad (9)$$

for functions $\alpha_{i,m} : \Omega \rightarrow \mathbb{R}_+$ and $\beta_{i,m} : \Omega \rightarrow \mathbb{R}_+$. These choices of multiplier yield the following SVO:

$$\begin{aligned} & \bar{\lambda}(\nu, \omega, a) \\ &= \min\{\Sigma a, K\} p(\Sigma a) \\ &+ \sum_{i,m} \left[\alpha_{i,m}(\omega_{i,m}) \mathbb{I}_{a_{i,m}=0} (u_i(\nu, \omega, (1, a_{i,-m}), a_{-i}) - (u_i(\nu, \omega, (0, a_{i,-m}), a_{-i}))) \right. \\ &\quad \left. + \beta_{i,m}(\omega_{i,m}) \mathbb{I}_{a_{i,m}=1} (u_i(\nu, \omega, (0, a_{i,-m}), a_{-i}) - (u_i(\nu, \omega, (1, a_{i,-m}), a_{-i}))) \right]. \end{aligned} \quad (10)$$

Lemma 2. *We have*

$$|r(\Sigma a) - q_i(a)| \leq \frac{M + M^2}{K}$$

for every $a \in \{0, \dots, M\}^N$.

Proof. Let $n = \Sigma a$. If $n \leq K$, then $r(\Sigma a) = q_i(a) = 1$. So suppose that $n > K$. Fixing $a = (a_1, \dots, a_N)$, let J be the set of agents with non-zero actions.

The rationing rule randomly orders the agents in J and serves them sequentially (filling their entire market order) until the good runs out, meaning that another agent cannot be served without allocating more than K units.

Let E_i be the set of orders of J in which i gets served (so that the sum of prior actions is less than $K - a_i$), and let F_i be the set of orders in which i is served when less than $K - M$ of the good has been allocated. Let q_i be the likelihood of E_i and π_i be the likelihood of F_i .

Claim 1: $q_i \geq \pi_j$ for all i and j in J . For an order ζ , let $\tilde{\zeta}$ denote the order obtained from ζ by swapping i and j . Note that this swap operation is a bijection between orders. Let $\Sigma_i(\zeta)$ be the sum of the actions up to and including agent i according to the order ζ . Then $\zeta \in E_i$ if and only if $\Sigma_i(\zeta) \leq K$, and $\zeta \in F_i$ if and only if $\Sigma_i(\zeta) - a_i \leq K - M$.

Suppose $\zeta \in F_j$. There are two possibilities: If i comes before j under ζ , then $\Sigma_j(\zeta) = \Sigma_i(\tilde{\zeta})$ (since both of those sums include i and j being served, as well as every other agent before j in ζ). Since $\zeta \in F_j$, we have $\Sigma_i(\tilde{\zeta}) = \Sigma_j(\zeta) \leq K - M + a_j \leq K$. Hence, $\tilde{\zeta} \in E_i$.

Now suppose i comes after j under ζ . Then the agents before i under $\tilde{\zeta}$ are the same as those before j under ζ , and hence $\Sigma_j(\zeta) - a_j = \Sigma_i(\tilde{\zeta}) - a_i$. This implies that $\Sigma_i(\tilde{\zeta}) = \Sigma_j(\zeta) - a_j + a_i \leq K - M + a_i \leq K$. Hence, $\tilde{\zeta} \in E_i$.

We have shown that for each equivalence class $(\zeta, \tilde{\zeta})$ under the i - j swap bijection, if $\zeta \in F_j$, then $\tilde{\zeta} \in E_i$. Hence, the number of elements of E_i must be at least that of F_j , so that $q_i \geq \pi_j$, as asserted by Claim 1.

Claim 2: $q_i - \pi_i \leq M^2/n$. To see why, fix the order in which agents in $J \setminus \{i\}$ are served. Note that agent i is equally likely to be anywhere in the J positions between members of $J \setminus \{i\}$. Now, I claim that there are at most M positions of agent i in which they are served but not served before $K - M$ units have been allocated. The number of such positions is maximized when the last M agents served (including i) all demand a single unit. At the same time, $|J| \geq n/M$ (since each agent in i demands at most M units). Hence, conditioning on the order in $J \setminus \{i\}$, the likelihood that i is served but not before $K - M$ units are allocated is at most $M/(n/M) = M^2/n$.

Claim 3: $K - M \leq \sum_i a_i q_i \leq K$. This is simply a statement about the expected number of units allocated, which is always between $K - M$ and K .

Claim 4: $q_i \geq (K - M)/n - M^2/n$. For any i and j , Claims 1 and 2 imply that

$$q_i \leq \pi_i + M^2/n \leq q_j + M^2/n$$

Hence, for any j ,

$$\sum_i a_i q_i \leq n(q_j + M^2/n),$$

so from Claim 3 we have

$$q_j \geq \frac{K - M}{n} - M^2/n.$$

Claim 5: $q_i \leq K/n + M^2/n$. From Claims 1 and 2,

$$q_i \geq \pi_j \geq q_j - M^2/n.$$

Now from Claim 3,

$$K \geq \sum_i a_i q_i \geq n(q_j - M^2/n) \implies q_j \leq \frac{K}{n} + \frac{M^2}{n}.$$

□

Thus, when K is large, we can approximate the utility as if the rationing rule is given by $r(\Sigma a)$,

$$\tilde{u}_i(\nu, \omega, a) = \sum_m a_{i,m} r(\Sigma a) (v(\nu, \omega_{i,m}) - p(\Sigma a)).$$

since Lemma 2 and the fact that v and p are bounded imply

$$|\tilde{u}_i(\nu, \omega, a) - u_i(\nu, \omega, a)| \leq \frac{B}{K}$$

for some constant $B > 0$. Thus, we have the following lower bound on the SVO

$$\begin{aligned} & \bar{\lambda}(\nu, \omega, a) \\ & \geq \min\{\Sigma a, K\} p(\Sigma a) \\ & + \sum_{i,m} \left[\alpha_{i,m}(\omega_{i,m}) \mathbb{I}_{a_{i,m}=0} (\tilde{u}_i(\nu, \omega, (1, a_{i,-m}), a_{-i}) - (\tilde{u}_i(\nu, \omega, (0, a_{i,-m}), a_{-i})) \right. \\ & \quad \left. + \beta_{i,m}(\omega_{i,m}) \mathbb{I}_{a_{i,m}=1} (\tilde{u}_i(\nu, \omega, (0, a_{i,-m}), a_{-i}) - (\tilde{u}_i(\nu, \omega, (1, a_{i,-m}), a_{-i})) \right] \\ & - \max_{i,m} (\alpha_{i,m}(\omega_{i,m}), \beta_{i,m}(\omega_{i,m})) NM(4B/K). \end{aligned} \tag{11}$$

Next, note that

$$|r(n+1) - r(n)| = \left| \min \left\{ \frac{K}{n+1}, 1 \right\} - \min \left\{ \frac{K}{n}, 1 \right\} \right| \leq \frac{1}{K},$$

and by definition, $|p(n+1) - p(n)| \leq \gamma$, where γ is the price impact.

Setting $n = \sum_{j \neq i,m} a_{j,m} + \sum_{m' \neq m} a_{i,m'}$, we have¹³

$$\begin{aligned} & \tilde{u}_i(\nu, \omega, (1, a_{i,-m}), a_{-i}) - \tilde{u}_i(\nu, \omega, (0, a_{i,-m}), a_{-i}) \\ & = r(n+1)(v(\nu, \omega_{i,m}) - p(n+1)) \\ & + \underbrace{\sum_{m' \neq m} a_{i,m'} (r(n+1)(v(\nu, \omega_{i,m'}) - p(n+1)) - r(n)(v(\nu, \omega_{i,m'}) - p(n)))}_{|\cdot| \leq M(\bar{v}/K + \gamma + \mathcal{C}/K)}. \end{aligned}$$

Combining with (11) then gives

$$\begin{aligned} & \bar{\lambda}(\nu, \omega, a) \geq \min\{\Sigma a, K\} p(\Sigma a) \\ & + \sum_{i,m} \left[\alpha_{i,m}(\omega_{i,m}) \mathbb{I}_{a_{i,m}=0} (r(\Sigma a + 1)(v(\nu, \omega_{i,m}) - p(\Sigma a + 1))) \right] \end{aligned}$$

¹³We use the fact that $|r(n+1)p(n+1) - r(n)p(n)| \leq |r(n+1)p(n+1) - r(n+1)p(n)| + |r(n+1)p(n) - r(n)p(n)| \leq \gamma + \mathcal{C}/K$.

$$\begin{aligned}
& - \beta_{i,m}(\omega_{i,m}) \mathbb{I}_{a_{i,m}=1} (r(\Sigma a)(v(\nu, \omega_{i,m}) - p(\Sigma a))) \Big] \\
& - \max_{i,m} (\alpha_{i,m}(\omega_{i,m}), \beta_{i,m}(\omega_{i,m})) NM(4B/K + \bar{v}/K + \gamma + \mathcal{C}/K).
\end{aligned}$$

By Lemma 1, this proves Proposition 1 with $C = 4B + \bar{v} + \mathcal{C}$.

A.2 Proof of Theorem 1

Let $K_N \approx \kappa N$, let $x \leq \kappa$ and $x \in (0, 1)$. Let (p_N) be a sequence of pricing rules with corresponding spans (p, x, ϵ_N) and price impacts γ_N , where $\epsilon_N \rightarrow 0$ and $\gamma_N \rightarrow 0$.

Let (δ_N) be a sequence converging to zero for which¹⁴

$$\mu_N \left(\sup_{z \in [\underline{\omega}, \bar{\omega}]} |F_N(z; \omega) - F(z)| > \delta_N \right) \leq \delta_N. \quad (12)$$

In light of Proposition 1, we work with the SVO in the separated economy (cf. Section 3.1):

$$\begin{aligned}
\lambda_N(\nu, \omega, a) = \min(\Sigma a, K_N) p_N(\Sigma a) + \sum_i [(1 - a_i) \alpha_i(\omega_i) (v(\nu, \omega_i) - p_N(\Sigma a + 1)) r(\Sigma a + 1) \\
- a_i \beta_i(\omega_i) (v(\nu, \omega_i) - p_N(\Sigma a)) r(\Sigma a)].
\end{aligned}$$

To simplify the notation, we have set $M = 1$ and $\omega_i = \omega_{i,1}$; this is without loss of generality, since in the separated economy each account is a distinct buyer with unit demand.

Let us define

$$\begin{aligned}
\tilde{\lambda}_N(\nu, \omega, a) \equiv \min(\Sigma a, K_N) p_N(\Sigma a) + \sum_i [(1 - a_i) \alpha_i(\omega_i) (v(\nu, \omega_i) - p_N(\Sigma a)) r(\Sigma a) \\
- a_i \beta_i(\omega_i) (v(\nu, \omega_i) - p_N(\Sigma a)) r(\Sigma a)]
\end{aligned}$$

Clearly,

$$\lambda_N(\nu, \omega, a) \geq \tilde{\lambda}_N(\nu, \omega, a) - N \max_i (\alpha_i(\omega_i)) (\bar{v}/K_N + \gamma_N + \mathcal{C}/K_N).$$

So for lower bounding $\lambda_N(\nu, \omega, a)/Nx$ we can focus on lower bounding $\tilde{\lambda}_N(\nu, \omega, a)/Nx$ and ignore the second term on the righthand side of the preceding inequality.

¹⁴ Let (δ_k) be a sequence converging to 0. By condition (3), for each k , there exists a N_k such that $\mu_N \left(\sup_{z \in [\underline{\omega}, \bar{\omega}]} |F_N(z; \omega) - F(z)| > \delta_k \right) \leq \delta_k$ for all $N \geq N_k$. Without loss, we can choose N_k to be strictly increasing in k . Then (12) is satisfied for $\delta_N = \delta_k$ where $N_k \leq N < N_{k+1}$.

Let $y = F^{-1}(1 - x)$. For a fixed $\rho > 0$, we choose the following multipliers:

$$\begin{aligned}\alpha_i(\omega_i) &= \frac{x}{F(y) - F(y - \rho)}, \beta_i(\omega_i) = 0, \text{ if } \omega_i > y - \rho; \\ \alpha_i(\omega_i) &= 0, \beta_i(\omega_i) = \frac{x}{F(y) - F(y - \rho)}, \text{ if } \omega_i \leq y - \rho.\end{aligned}\tag{13}$$

We will show that there exists a sequence (ζ_N) converging to zero such that

$$\frac{\tilde{\lambda}_N(\nu, \omega, a)}{Nx} \geq v(\nu, y - \rho) - \zeta_N,\tag{14}$$

for every N , a , ν , and ω satisfying

$$\sup_{z \in [\underline{\omega}, \bar{\omega}]} |F_N(z; \omega) - F(z)| \leq \delta_N.\tag{15}$$

Because of (12), as $N \rightarrow \infty$ we can ignore the integral of $\frac{\tilde{\lambda}_N(\nu, \omega, a)}{Nx}$ over ω that fails condition (15). Proposition 1 (when applying Proposition 1 we revert back to the notation of NM buyers with unit demand and values $v(\nu, \omega_{i,m})$) then implies that

$$\liminf_{N \rightarrow \infty} \left(\frac{R(p_N, \mu_N)}{Nx} - \int_{\nu, \omega} v(\nu, y - \rho) \mu_N(d\nu, d\omega) \right) \geq 0.$$

Sending $\rho \rightarrow 0$ then establishes the revenue guarantee of the market order mechanism is asymptotically at least $Nx \int_{\nu, \omega} v(\nu, y) \mu_N(d\nu, d\omega)$. In this step, it is important that we first take $N \rightarrow \infty$ and then $\rho \rightarrow 0$. This ensures that the Lagrange multipliers in (6) are bounded, so that the lower bound in Proposition 1 is applicable.

Case 1:

Consider $n \in [N(x - \epsilon_N - \delta_N), N(x + \epsilon_N)]$. Because $\alpha_i(\omega_i) + \beta_i(\omega_i)$ is a constant, subject to $\sum a = n$, $\tilde{\lambda}_N(\nu, \omega, a)$ is clearly minimized by a where $a_i = 1$ for the n highest $v(\nu, \omega_i)$. By rearrangement we can assume

$$\omega_1 \geq \omega_2 \geq \dots \geq \omega_N,$$

and thus

$$\begin{aligned}\min_{a: \sum a = n} \tilde{\lambda}_N(\nu, \omega, a) &= \min(n, K_N) p_N(n) - \sum_{i=1}^n \beta_i(\omega_i) (v(\nu, \omega_i) - p_N(n)) r(n) \\ &\quad + \sum_{i=n+1}^N \alpha_i(\omega_i) (v(\nu, \omega_i) - p_N(n)) r(n).\end{aligned}$$

Let n' be the largest i such that $\omega_i > y - \rho$, i.e., the largest i such that $\alpha_i(\omega_i) > 0$ and $\beta_i(\omega_i) = 0$. Since by definition $F_N(y - \rho; \omega) = 1 - n'/N$, condition (15) implies the following range for n' :

$$N(1 - F(y - \rho) - \delta_N) \leq n' \leq N(1 - F(y - \rho) + \delta_N). \quad (16)$$

Since $F(y) = 1 - x > F(y - \rho)$, when N is sufficiently large $n' \geq N(1 - F(y - \rho) - \delta_N) > N(x + \epsilon_N) \geq n$. Substituting the multipliers from (13), we arrive at

$$\begin{aligned} \min_{a: \Sigma a = n} \tilde{\lambda}_N(\nu, \omega, a) &= \min(n, K_N) p_N(n) + \sum_{i=n+1}^{n'} \frac{x}{F(y) - F(y - \rho)} (v(\nu, \omega_i) - p_N(n)) r(n) \\ &\geq \min(n, K_N) p_N(n) + \sum_{i=n+1}^{n'} \frac{x}{F(y) - F(y - \rho)} (v(\nu, y - \rho) - p_N(n)) r(n) \\ &\geq N(x - \epsilon_N - \delta_N) p_N(n) + \frac{(n' - n)x}{F(y) - F(y - \rho)} v(\nu, y - \rho) \frac{\kappa}{\kappa + \epsilon_N} \\ &\quad - \frac{(n' - n)x}{F(y) - F(y - \rho)} p_N(n), \end{aligned}$$

where we used the fact that $n \in [N(x - \epsilon_N - \delta_N), N(x + \epsilon_N)]$ and $\frac{\kappa N}{(\kappa + \epsilon_N)N} \leq r(n) \leq 1$ (since $x \leq \kappa = K_N/N$). In the last expression above, the terms involving $p_N(n)$ essentially cancel out, and the coefficient of $v(\nu, y - \rho)$ is approximately 1. In other words, the above equations imply that (14) holds for $\Sigma a = n$ with

$$\begin{aligned} \zeta_N = \sup_{\substack{n \in [N(x - \epsilon_N - \delta_N), N(x + \epsilon_N)], \\ n' \in [N(1 - F(y - \rho) - \delta_N), N(1 - F(y - \rho) + \delta_N)]}} &\left| \frac{x - \epsilon_N - \delta_N}{x} - \frac{(n' - n)}{N(F(y) - F(y - \rho))} \right| \mathcal{C} \\ &+ \left| \frac{n' - n}{N(F(y) - F(y - \rho))} \frac{\kappa}{\kappa + \epsilon_N} - 1 \right| \bar{v}. \end{aligned}$$

Clearly, $\zeta_N \rightarrow 0$ since $\epsilon_N \rightarrow 0$, $\delta_N \rightarrow 0$, and $x = 1 - F(y)$.

For the remaining two cases, define

$$\begin{aligned} \tilde{\lambda}_N(\nu, \omega, a; \tilde{p}) &= \min(\Sigma a, K_N) \tilde{p} + \sum_i [(1 - a_i) \alpha_i(\omega_i) (v(\nu, \omega_i) - \tilde{p}) r(\Sigma a) \\ &\quad - a_i \beta_i(\omega_i) (v(\nu, \omega_i) - \tilde{p}) r(\Sigma a)], \end{aligned}$$

for a fixed $\tilde{p} \in [0, \mathcal{C}]$.

Case 2:

Consider an $a \in \{0, 1\}^N$ such that $\Sigma a < N(x - \epsilon_N - \delta_N)$. Set $\tilde{p} = p_N(\Sigma a)$. Since $\Sigma a < N(x - \epsilon_N)$, we have $\tilde{p} \leq \underline{p}$. We claim that there exists an $\tilde{a} \in \{0, 1\}^N$ such that $\Sigma \tilde{a} \geq N(x - \epsilon_N - \delta_N)$ and

$$\tilde{\lambda}_N(\nu, \omega, a; \tilde{p}) \geq \tilde{\lambda}_N(\nu, \omega, \tilde{a}; \tilde{p}).$$

Then (14) holds for $\Sigma a < N(x - \epsilon_N - \delta_N)$ since we have shown for case 1 that $\frac{\tilde{\lambda}_N(\nu, \omega, \tilde{a}; \tilde{p})}{Nx} \geq v(\nu, y - \rho) - \zeta_N$.

Since $F(y) = 1 - x$, $F_N(y; \omega) \leq 1 - x + \delta_N$, so there are at least $N(x - \delta_N)$ i 's such that $\omega_i > y$. Since $\Sigma a < N(x - \delta_N)$, then there exists an i such that $a_i = 0$ and $\omega_i > y$. Since $r(\Sigma a) = 1$, we can weakly lower $\tilde{\lambda}_N(\nu, \omega, a; \tilde{p})$ by finding an i such that $a_i = 0$ and $\omega_i > y$, and increasing its a_i from 0 to 1, if

$$\tilde{p} - (\alpha_i(\omega_i) + \beta_i(\omega_i))(v(\nu, \omega_i) - \tilde{p}) \leq 0,$$

Clearly, the above condition is satisfied by the multipliers in (13) when ρ is sufficiently small, since either $v(\nu, \omega_i) \geq v(\nu, y) > \underline{p} \geq \tilde{p}$, or $\underline{p} = \tilde{p} = 0$. We repeatedly apply this operation until $\Sigma a \geq N(x - \epsilon_N - \delta_N)$ and let the resulting a be $\tilde{a}$.

Case 3:

Consider an $a \in \{0, 1\}^N$ such that $\Sigma a > N(x + \epsilon_N)$. Let $\tilde{p} = p_N(\Sigma a)$. Since $\Sigma a > N(x + \epsilon_N)$, we have $\tilde{p} \geq \bar{v}$. We claim that there exists an $\tilde{a} \in \{0, 1\}^N$ such that $\Sigma \tilde{a} \leq N(x + \epsilon_N)$ and

$$\tilde{\lambda}_N(\nu, \omega, a; \tilde{p}) \geq \tilde{\lambda}_N(\nu, \omega, \tilde{a}; \tilde{p}).$$

As in Case 1, this implies that (14) holds for $\Sigma a > N(x + \epsilon_N)$.

This is clearly true when $\Sigma a \leq K_N$ so $r(\Sigma a) = 1$: we lower $\tilde{\lambda}_N(\nu, \omega, a; \tilde{p})$ by decreasing any $a_i = 1$ to $a_i = 0$. And when $\Sigma a > K_N$,

$$\begin{aligned} \tilde{\lambda}_N(\nu, \omega, a; \tilde{p}) &= K_N \tilde{p} - \sum_i \alpha_i(\omega_i)(\tilde{p} - v(\nu, \omega_i)) \frac{K_N}{\Sigma a} \\ &\quad + \sum_i a_i (\alpha_i(\omega_i) + \beta_i(\omega_i))(\tilde{p} - v(\nu, \omega_i)) \frac{K_N}{\Sigma a} \end{aligned}$$

We lower $\tilde{\lambda}_N(\nu, \omega, a; \tilde{p})$ by choosing $\bar{i}$ whose $\omega_{\bar{i}}$ is lowest among all i such that $a_i = 1$, and decreasing $a_{\bar{i}}$ from 1 to 0. We repeatedly apply this operation until $\Sigma a \leq N(x + \epsilon_N)$ and let the resulting a be $\tilde{a}$.

Finally, we show that

$$\limsup_{N \rightarrow \infty} \left(\frac{R(p_N, \mu_N)}{Nx} - \int_{\nu, \omega} v(\nu, y) \mu_N(d\nu, d\omega) \right) \leq 0. \quad (17)$$

For each N , consider an information structure where every buyer i knows everyone's idiosyncratic component $\omega = (\omega_1, \dots, \omega_N)$ but has no information about ν beyond the prior μ_N (so $s_i = \omega_{-i}$). As before, it is without loss to focus on ω where (15) is satisfied. By rearrangement, we can assume

$$\omega_1 \geq \omega_2 \geq \dots \geq \omega_N.$$

Define

$$v_i(\omega) \equiv \int_{\nu} v(\nu, \omega_i) \mu_N(d\nu \mid \omega).$$

Let n be the largest i such that $v_i(\omega) \geq p_N(i)$ and $i \leq (x + \epsilon_N)N$.

Then it is an equilibrium for buyer i to place an order ($a_i = 1$) if $i \leq n$ and to not place an order ($a_i = 0$) if $i > n$. The revenue in this equilibrium is

$$np_N(n) \leq nv_n.$$

Because of the condition on the span of p_N and condition (15), we have

$$(x - \epsilon_N - \delta_N)N \leq n \leq (x + \epsilon_N)N.$$

Since $\epsilon_N \rightarrow 0$ and $\delta_N \rightarrow 0$, (17) follows.

A.3 Proof of Theorem 2

Let (μ_N) be a sequence of priors satisfying condition (12) for some sequence (δ_N) converging to zero (see footnote 14), and let (p_N) be a sequence of pricing rules with price impact γ_N tending to zero. Let f be the probability density function for the limit CDF F .

Because of (12), we can focus on ω such that

$$\sup_y |F_N(y; \omega) - F(y)| \leq \delta_N.$$

which implies for a continuously differentiable function $g : [z, \bar{\omega}] \rightarrow \mathbb{R}$,

$$\begin{aligned} & \left| \frac{1}{NM} \sum_{i,m} \mathbb{I}_{\omega_{i,m} \geq z} g(\omega_{i,m}) - \int_{y=z}^{\bar{\omega}} g(y) f(y) dy \right| \\ &= \left| (1 - F_N(z; \omega)) g(\underline{\omega}) + \int_{y=z}^{\bar{\omega}} g'(y) (1 - F_N(y; \omega)) dy - (1 - F(z)) g(z) - \int_{y=z}^{\bar{\omega}} g'(y) (1 - F(y)) dy \right| \\ &\leq \delta_N \max_{y \in [z, \bar{\omega}]} |g(y)| + \delta_N (\bar{\omega} - \underline{\omega}) \max_{y \in [z, \bar{\omega}]} |g'(y)|. \end{aligned} \tag{18}$$

Let n_N^* be the maximizer for $W^*(\mu_N, p_N)$ in (7). Let $y_N^* = F^{-1}(1 - n_N^*/(NM))$. By definition, we have $F(y_N^*) = 1 - n_N^*/(NM)$. We set

$$\alpha_{N,i,m}(\omega_{i,m}) = \mathbb{I}_{\omega_{i,m} \geq y_N^*}, \quad \beta_{N,i,m}(\omega_{i,m}) = 1 - \alpha_{N,i,m}(\omega_{i,m}).$$

Define the following SVO with above Lagrange multipliers, where we assume $a_i \in \{0, 1\}^M$ as in Section 3.1:

$$\begin{aligned} \lambda_N(\nu, \omega, a) & \\ = \sum_{i,m} a_{i,m} v(\nu, \omega_{i,m}) - \sum_{l \leq \Sigma a} p_N(l) & \\ + \sum_{i,m} \left(\mathbb{I}_{a_{i,m}=0} \alpha_{N,i,m}(\omega_{i,m}) (v(\nu, \omega_{i,m}) - p_N(\Sigma a + 1)) - \mathbb{I}_{a_{i,m}=1} \beta_{N,i,m}(\omega_{i,m}) (v(\nu, \omega_{i,m}) - p_N(\Sigma a)) \right) & \end{aligned} \quad (19)$$

This is the analogue of the SVO in (6) for the separated economy, except that the revenue objective has been replaced with social welfare. By exactly the same logic as in Proposition 1, we have the following lower bound on the welfare guarantee:¹⁵

$$\underline{W}(p_N, \mu_N) \geq \int_{\nu, \omega} \min_a \lambda_N(\nu, \omega, a) \mu_N(d\nu, d\omega) - NM\gamma_N,$$

where $\gamma_N \rightarrow 0$ as $N \rightarrow \infty$. The rest of the proof proceeds by lower bounding $\lambda_N(\nu, \omega, a)$. We have

$$\begin{aligned} \lambda_N(\nu, \omega, a) & \\ = \sum_{i,m} a_{i,m} v(\nu, \omega_i) - \sum_{l \leq \Sigma a} p_N(l) & \\ + \sum_{i,m} \left(\mathbb{I}_{a_{i,m}=0} \alpha_{N,i,m}(\omega_i) (v(\nu, \omega_i) - p_N(\Sigma a)) - \mathbb{I}_{a_{i,m}=1} \beta_{N,i,m}(\omega_i) (v(\nu, \omega_{i,m}) - p_N(\Sigma a)) \right) & \\ = \Sigma a p_N(\Sigma a) - \sum_{l \leq \Sigma a} p_N(l) & \\ + \sum_{i,m} \left(\mathbb{I}_{a_{i,m}=0} \alpha_{N,i,m}(\omega_{i,m}) (v(\nu, \omega_{i,m}) - p_N(\Sigma a)) + \mathbb{I}_{a_{i,m}=1} \alpha_{N,i,m}(\omega_{i,m}) (v(\nu, \omega_{i,m}) - p_N(\Sigma a)) \right) & \\ = \Sigma a p_N(\Sigma a) - \sum_{l \leq \Sigma a} p_N(l) + \sum_{i,m} \mathbb{I}_{\omega_{i,m} \geq y_N^*} (v(\nu, \omega_{i,m}) - p_N(\Sigma a)). & \end{aligned}$$

¹⁵Note that the term $\max_{i,m,\omega_{i,m}} (\alpha_{N,i,m}(\omega_{i,m}), \beta_{N,i,m}(\omega_{i,m}))$ in Proposition 1 is equal to one, and we have $q_i(a) = r(\Sigma a) = 1$, which results in $C = 0$ in Proposition 1.

Applying (18), we then have

$$\begin{aligned}\lambda_N(\nu, \omega, a) &\geq \Sigma a p_N(\Sigma a) - \sum_{l \leq \Sigma a} p_N(l) + NM \int_{y \geq y_N^*} (v(\nu, y) - p_N(\Sigma a)) f(y) dy - NM \delta_N C \\ &= (\Sigma a - n_N^*) p_N(\Sigma a) - \sum_{l \leq \Sigma a} p_N(l) + NM \int_{y \geq y_N^*} v(\nu, y) f(y) dy - NM \delta_N C, \quad (20)\end{aligned}$$

where

$$C = (\bar{v} + \mathcal{C}) + (\bar{\omega} - \underline{\omega}) D, \quad D = \max_{y \in [\underline{\omega}, \bar{\omega}]} \left| \frac{\partial v}{\partial y}(\nu, y) \right|.$$

Now, define the function

$$J(n) \equiv (n - n_N^*) p_N(n) - \sum_{l \leq n} p_N(l).$$

Then (20) is equal to

$$J(\Sigma a) + NM \int_{y \geq y_N^*} v(\nu, y) f(y) dy - NM \delta_N C.$$

Now,

$$\begin{aligned}J(n+1) - J(n) &= (n+1 - n^*) p_N(n+1) - (n - n^*) p_N(n) - p_N(n+1) \\ &= (n - n^*) (p_N(n+1) - p_N(n)).\end{aligned}$$

Because p_N is non-decreasing, J is single-troughed at $n = n^*$, and (20) is at least

$$\begin{aligned}& - \sum_{l \leq n_N^*} p_N(l) + NM \int_{y \geq y_N^*} v(\nu, y) f(y) dy - NM \delta_N C \\ & \geq - \sum_{l \leq n_N^*} p_N(l) + \sum_{i, m} \mathbb{I}_{\omega_{i, m} \geq y_N^*} v(\nu, \omega_{i, m}) - 2NM \delta_N C \\ & \geq - \sum_{l \leq n_N^*} p_N(l) + \sum_{l=1}^{n_N^*} v(\nu, \omega^{(l)}) - NM \delta_N \bar{v} - 2NM \delta_N C,\end{aligned}$$

where the last line is due to $F(y_N^*) = 1 - n_N^*/NM$ and $|F(y) - F_N(y)| \leq \delta_N$ for all $y < y_N^*$, so the number of (i, m) with $\omega_{i, m} \geq y_N^*$ is between $n_N^* - NM \delta_N$ and $n_N^* + NM \delta_N$. Thus, the theorem follows from the analogue of Proposition 1 for welfare.

References

- BALI, V. AND M. JACKSON (2002): “Asymptotic Revenue Equivalence in Auctions,” *Journal of Economic Theory*, 106, 161–176.
- BARELLI, P., S. GOVINDAN, AND R. WILSON (2024): “Strategic Foundations of Efficient Rational Expectations,” *The Review of Economic Studies*, 91, 3190–3216.
- BEJA, A. (1977): “The limits of price information in market processes,” Working paper, University of California at Berkeley.
- BERGEMANN, D., B. BROOKS, AND S. MORRIS (2017): “First-Price Auctions with General Information Structures: Implications for Bidding and Revenue,” *Econometrica*, 85, 107–143.
- BERGEMANN, D. AND S. MORRIS (2016): “Bayes Correlated Equilibrium and the Comparison of Information Structures in Games,” *Theoretical Economics*, 11, 487–522.
- BROOKS, B. AND S. DU (2021): “Optimal Auction Design With Common Values: An Informationally Robust Approach,” *Econometrica*, 89, 1313–1360.
- (2024): “On the Structure of Informationally Robust Optimal Mechanisms,” *Econometrica*, 92, 1391–1438.
- BROOKS, B., S. DU, AND A. HABERMAN (2025): “Robust Predictions with Bounded Information,” Working paper, University of Chicago, University of California-San Diego, and Stanford University.
- CRÉMER, J. AND R. P. MCLEAN (1988): “Full Extraction of the Surplus in Bayesian and Dominant Strategy Auctions,” *Econometrica*, 1247–1257.
- DU, S. (2018): “Robust Mechanisms Under Common Valuation,” *Econometrica*, 86, 1569–1588.
- DU, S. AND H. ZHU (2017): “What is the Optimal Trading Frequency in Financial Markets?” *The Review of Economic Studies*, 84, 1606–1651.
- ENGELBRECHT-WIGGANS, R., P. R. MILGROM, AND R. J. WEBER (1983): “Competitive Bidding and Proprietary Information,” *Journal of Mathematical Economics*, 11, 161–169.
- HABERMAN, A. AND R. JAGADEESAN (2025): “Auctions with Withdrawal Rights: A Foundation for Uniform Price,” Working paper, Stanford University.

- HARTLINE, J., V. SYRGKANIS, AND E. TARDOS (2015): “No-regret learning in Bayesian games,” *Advances in Neural Information Processing Systems*, 28.
- JACKSON, M. O. (2009): “Non-existence of equilibrium in Vickrey, second-price, and English auctions,” *Review of Economic Design*, 13, 137–145.
- JACKSON, M. O. AND I. KREMER (2007): “On the informational inefficiency of discriminatory price auctions,” *Journal of Economic Theory*, 132, 507–517.
- KREMER, I. (2002): “Information Aggregation in Common Value Auctions,” *Econometrica*, 70, 1675–1682.
- KYLE, A. S. (1985): “Continuous Auctions and Insider Trading,” *Econometrica*, 53, 1315–1336.
- LAMBERT, N. S., M. OSTROVSKY, AND M. PANOV (2018): “Strategic Trading in Informationally Complex Environments,” *Econometrica*, 86, 1119–1157.
- LAUERMANN, S. AND A. SPEIT (2023): “Bidding in Common-Value Auctions With an Unknown Number of Competitors,” *Econometrica*, 91, 493–527.
- LAUERMANN, S. AND A. WOLINSKY (2017): “Bidder Solicitation, Adverse Selection, and the Failure of Competition,” *American Economic Review*, 107, 1399–1429.
- LOMYS, N. AND L. MAGNOLFI (2025): “Estimation of Games under No Regret: Structural Econometrics for AI,” Working paper, University of Naples Federico II and University of Wisconsin-Madison.
- MILGROM, P. R. (1979): “A Convergence Theorem for Competitive Bidding with Differential Information,” *Econometrica*, 679–688.
- (1981): “Rational expectations, information acquisition, and competitive bidding,” *Econometrica*, 921–943.
- MYERSON, R. B. (1981): “Optimal Auction Design,” *Mathematics of Operations Research*, 6, 58–73.
- OSTROVSKY, M. (2012): “Information Aggregation in Dynamic Markets with Strategic Traders,” *Econometrica*, 2595–2647.
- PESENDORFER, W. AND J. M. SWINKELS (1997): “The Loser’s Curse and Information Aggregation in Common Value Auctions,” *Econometrica*, 65, 1247–1281.

- (2000): “Efficiency and Information Aggregation in Auctions,” *American Economic Review*, 90, 499–525.
- RENY, P. J. AND M. PERRY (2006): “Toward a strategic foundation for rational expectations equilibrium,” *Econometrica*, 74, 1231–1269.
- WILSON, R. (1977): “A Bidding Model of Perfect Competition,” *The Review of Economic Studies*, 511–518.